

INTERMEDIATE COURSE

GROUP – II

REVISION TEST PAPERS

SEPTEMBER, 2024



BOARD OF STUDIES

THE INSTITUTE OF CHARTERED ACCOUNTANTS OF INDIA

(Set up by an Act of Parliament)

New Delhi

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REVISION TEST PAPER, SEPTEMBER, 2024 – OBJECTIVE & APPROACH

(Students are advised to go through the following paragraphs carefully to derive maximum benefit out of this RTP)

I. Objective of Revision Test Paper

Revision Test Papers are one among the many educational inputs provided by the Board of Studies (BOS) to its students. Popularly referred to as RTP by the students, it is one of the very old publications of the BOS whose significance and relevance from the examination perspective has stood the test of time.

The primary objectives of the RTP are:

- To help students get an insight of their preparedness for the forthcoming examination;
- To update them on the latest developments relevant for the forthcoming examination in select subjects;
- To enhance the confidence level of the students adequately.

Students must bear in mind that the RTP contains a variety of questions based on different topics of the syllabi and thus a comprehensive study of the entire syllabus is a pre-requisite before answering the questions of the RTP. In other words, in order to derive maximum benefit out of the RTPs, it is advised that before proceeding to solve the questions given in the RTP, students ought to have thoroughly read the Study Materials and Statutory Update, wherever applicable.

The topics on which the questions are set herein have been carefully selected and meticulous attention has been paid in framing different types of questions. Detailed answers are provided to enable the students to do a self-assessment and have a focused approach for effective preparation.

Live Virtual Classes by renowned subject experts conducted free of charge for the students of Foundation, Intermediate and Final levels provide the students much required support in preparing for their exams conveniently at home as these classes can be accessed live or viewed later as recorded lectures through hand-held devices such as smart

phones, laptops, I-pads, tablets, etc. anytime anywhere. Further, students are advised to attempt the Multiple-Choice Questions (MCQs) at MCQ Paper Practice Portal which is a holistic platform for self-assessment within the stipulated timeframe.

Students are welcome to send their suggestions for fine tuning the RTP to the Joint Director, Board of Studies, The Institute of Chartered Accountants of India, A-29, Sector-62, Noida 201309 (Uttar Pradesh). RTP is also available on BOS Knowledge Portal at <https://boslive.icai.org> for downloading.

II. Planning and preparing for examination

Ideally, when the RTP reaches your hand, you must have finished reading the relevant Study Materials of all the subjects available at the BoS Knowledge Portal. Get a good grasp of the concepts/ provisions/ amendments/ cases discussed therein.

After reading the Study Materials alongwith Statutory Update thoroughly, then, proceed to solve the questions given in the RTP on your own. RTP is an effective tool to revise and refresh the concepts and provisions discussed in the Study Material. RTPs are provided to you to help you assess your level of preparation. Hence you must solve the questions given therein on your own and thereafter compare your answers with the answers given therein.

Examination tips

How well a student fares in the examination depends upon the level and depth of his preparation. However, there are certain important points which can help a student better his performance in the examination. These useful tips are given below:

- Reach the examination hall well in time.
- As soon as you get the question paper, read it carefully and thoroughly. You are given separate 15 minutes for reading the question paper.
- Plan your time so that appropriate time is awarded for each question.

- First impression is the last impression. The question which you can answer in the best manner should be attempted first.
- Always attempt to do all questions. Therefore, it is important that you must finish each question within allocated time. Keep sometime for checking the answers as well.
- Read the question carefully more than once before starting the answer to understand very clearly as to what is required.
- Answer all parts of a question one after the other; do not answer different parts of the same question at different places.
- Write in a neat and legible hand-writing.
- Always be concise and write to the point and do not try to fill pages unnecessarily.
- There must be logical expression of the answer.
- In case a question is not clear, you may state your assumptions and then answer the question.
- Check your answers carefully and underline important points before leaving the examination hall.
- In case of case scenario based MCQs, read the facts given in the case attentively. Also, read each MCQ based thereon and all the options carefully, before choosing the correct answer.

III. Subject-wise Applicability

PAPER – 4: COST AND MANAGEMENT ACCOUNTING

The Revision Test Paper (RTP) of Cost and Management Accounting comprises of two Divisions of total 19 questions: Division A (Case scenario and caselet based MCQs) which contains 7 case scenarios and caselets and Division B (Descriptive questions) which contains 12 questions for full coverage of the syllabus. Theoretical questions along with computational problems have also been incorporated so that you are able to give emphasis to the theoretical portion of the syllabus as well. Since this paper's inclination is more towards numerical-oriented questions which involve mathematical calculations, therefore, it is very important that you have thoroughly studied the theoretical aspects of

the subject and are also clear about the concepts and logic behind the mathematical workings and formulae.

A summary of the questions both theoretical and computational has been given for your reference:

Q. No.	Topic	About the Problem
Division A		
1.	Material Cost	Calculation of material cost.
2.	Standard Costing	Calculation of different variances.
3.	Overheads	Calculation of under/over absorbed overheads.
4.	Process Costing	Calculation of abnormal gain.
5.	Service Costing	Calculation of room rent.
6.	Service Costing	Calculation of cost per rupee of insured value.
7.	Budget and Budgetary	Calculation of production budget.
Division B		
8.	Employee Cost	Calculation of labour turnover.
9.	Overheads	Calculation of machine hour rate
10.	Activity Based Costing (ABC) Method	Calculation of product cost.
11.	Cost Sheet	Preparation of Statement of Cost.
12.	Cost Accounting System	Preparation of Reconciliation.
13.	Job and Batch Costing	Compute the cost and profit/loss per batch and overall position.
14.	Process Costing	Preparation of Statement of equivalent production, cost per unit and cost of units transferred.
15.	Joint product and By product	Calculation of joint cost and profit.

16.	Service Costing	Calculation of toll costing and toll fee.
17.	Marginal Costing	Calculation of break-even sales and rate of return.
18.	Budget and Budgetary Control	Preparation of production budget and material purchase budget.
19(a).	Joint product and By product	Apportionment of joint costs by NRV method.
19(b).	Service Costing	Service costing & product costing.
19(c).	Introduction to Cost and Management Accounting	Controllable & uncontrollable variances.
19(d).	Introduction to Cost and Management Accounting	Standard & Discretionary cost.

PAPER – 5: AUDITING AND ETHICS

The Revisionary Test Paper (RTP) is a tool to refresh your knowledge which you have acquired while doing your conceptual study from Study Material and other modes of knowledge like BoS Live Virtual Classes, student journal, bare acts etc.

RTP of Auditing and Ethics for September 2024 comprises twenty questions, including case scenario-based multiple-choice questions, independent multiple choice questions, and descriptive questions. These questions cover the entire syllabus, which is divided into eleven chapters as discussed in the study material.

These 20 questions are taken from different topics like Nature, Objective and Scope of Audit, Audit Strategy, Audit Planning and Audit Program, Risk Assessment and Internal Control, Audit Evidence, Audit of Items of Financial Statements, Audit Documentation, Completion and Review, Audit Report, Special Features of Audit of Different Type of Entities, Audit of Banks and Ethics and Terms of Audit Engagements. The chapter's name is clearly indicated before each question. The questions in the RTP have been arranged in the same sequence as prescribed in

the study material to facilitate easy revision by the students. An attempt has been made to cover the syllabus comprehensively.

PAPER – 6: FINANCIAL MANAGEMENT AND STRATEGIC MANAGEMENT

Section – A: Financial Management

The Revision Test Paper (RTP) of Financial Management comprises of two Divisions of total 12 questions: Division A (Case scenario based MCQs) which contains 1 case scenario and 2 caselets and Division B (Descriptive questions) which contains 9 questions for full coverage of the syllabus. Theoretical questions along with computational problems have also been incorporated so that you are able to give emphasis to the theoretical portion of the syllabus as well. Since this paper's inclination is more towards numerical-oriented questions which involve mathematical calculations, therefore, it is very important that you have thoroughly studied the theoretical aspects of the subject and are also clear about the concepts and logic behind the mathematical workings and formulae.

A summary of the questions both theoretical and computational has been given for your reference:

Q. No.	Topic	About the Problem
Division A		
1.	Investment Decision	Analysis for a project.
2.	Ratio Analysis	Creditors turnover ratio
3.	Leverage	Analysis of financial leverage
Division B		
4.	Ratio Analysis	Preparation of balance sheet.
5.	Cost of Capital	Calculation of WACC.
6.	Capital Structure	Calculation of market value, cost of capital, cost of equity.

7.	Capital Structure	Calculation of market value, cost of capital, equity capitalization rate using M.M. approach.
8.	Leverage	Calculation of leverages.
9.	Dividend Decision	Calculation of market price of the share using gordon's model and walter's model.
10.	Working Capital	Preparation of statement of working capital requirement.
11.	Working Capital	Evaluation of credit policies.
12(a).	Scope and Objectives of Financial Management	Advantages and disadvantages of Wealth maximisation v/s Profit maximisation.
12(b).	Types of Financing	Type of preference shares.

Section – B: Strategic Management

The Revision Test Paper on the subject of Strategic Management for the September 2024 examination contains sixteen questions. The questions have been selected from all the sections/chapters uniformly to cover whole syllabus. Questions are based on different skill levels, i.e., "Comprehension & Knowledge" as well as "Analysis & Application".

The questions included are of different categories – multiple choice questions based on case scenario and application based multiple choice questions, distinguish between, short notes, descriptive and questions based on practical scenarios. The first question contains multiple choice questions based on case scenario are subdivided into five different parts. Questions from two to six are application based multiple choice questions. All multiple-choice questions are given with four alternatives and the student has to opt for the correct option. Subsequently, ten different questions have been included to cover all the five chapters of the syllabus. Chapter names have been mentioned before questions. A descriptive question based on practical scenario has been included from each section. Another descriptive question has also been included from each section of the syllabus.

The students should take up this Revision test paper as a tool to check their preparedness in the subject. Mere reading of Revision Test Paper will not be helpful. To properly self-assess the preparation in the subject, students must attempt the questions on their own. Compare your answers with the suggested answers and hints given to assess the level of preparation and identify areas where more focus is required. Then you may work on these areas to improve the quality of answers that you write.

Work hard and perform well in the examination!



PAPER – 4: COST AND MANAGEMENT ACCOUNTING



QUESTIONS

Division A: Case Scenario

Material Cost

1. 'Axe Trade', an unregistered supplier under GST, purchased material from Vye Ltd. which is registered supplier under GST. During the month of June 2024, the Axe Traders has purchased a lot of 5,000 units on credit from Vye Ltd. The information related to the purchase are as follows:

Listed price of one lot of 5,000 units	- ₹ 2,50,000
Trade discount	- @ 10% on listed price
CGST and SGST (Credit available)	- 18% (9% CGST + 9% SGST)
Cash discount	- @ 10%

(Will be given only if payment is made within 30 days.)

Toll Tax paid	₹ 5,000
Freight and Insurance	₹ 17,220
Demurrage paid to transporter	₹ 5,000
Commission and brokerage on purchases	₹ 10,000
Amount deposited for returnable containers	₹ 30,000
Amount of refund on returning the container	₹ 20,000
Other Expenses	@ 2% of total cost

A 20% shortage in material on receipt is expected considering the nature of the raw material.

The payment to the supplier was made within 21 days of the purchases.

- (i) If Axe Traders pays the supplier within 30 days of purchase, then, what is the total amount of cash discount received from the supplier and how it is treated to calculate material cost?
 - (a) ₹ 25,000 & it will not be deducted from the material cost
 - (b) ₹ 26,550 & it will be deducted from the material cost
 - (c) ₹ 26,550 & it will not be deducted from the material cost
 - (d) ₹ 22,500 & it will not be deducted from the material cost
- (ii) What will be the amount of other expenses and how it is treated in material cost?
 - (a) ₹ 6,154.40 & it will be added with the material cost
 - (b) ₹ 6,280.00 & it will be added with the material cost
 - (c) ₹ 5,344.40 & it will be added with the material cost
 - (d) ₹ 5,453.47 & it will not be added with the material cost
- (iii) What is the amount of GST and how will it be treated in cost sheet of Axe Traders?
 - (a) ₹ 40,500 & it will not be added with material cost
 - (b) ₹ 40,500 & it will be added with material cost
 - (c) ₹ 45,000 & it will not be added with material cost
 - (d) ₹ 45,000 & it will be added with material cost
- (iv) What is the total material cost chargeable in the cost sheet of Axe Traders?
 - (a) ₹ 3,14,000
 - (b) ₹ 2,73,500
 - (c) ₹ 2,72,673
 - (d) ₹ 3,13,874

- (v) The number of good units and cost per unit of the materials received are:
- (a) 5,000 units & ₹ 62.80
 - (b) 5,000 units & ₹ 54.70
 - (c) 4,000 units & ₹ 78.50
 - (d) 4,000 units & ₹ 68.38

Standard Costing

2. ABC Pvt Ltd is engaged in the manufacture of a Product Q. The product has the following standard production requirements determined by the technical team of the company post satisfactory completion of test run.

Raw Material Z – 2 units @ ₹ 2 per unit

Skilled labour of – 2.5 hours @ ₹ 5 per hour

Fixed Overheads – ₹ 7.5 per unit

The input of Raw material Z has a yield of 80% everytime when infused into production. The actual quantity of Raw material Z consumed for production during the year was 24,000 units. The Usage variance of Material Z was 2,000 Favourable. Further the actual amount of material cost for the material consumed amounted to ₹ 45,000.

During the said year, the actual working hours were 30,000 for which the labour cost paid by the company amounted to ₹1,20,000. The idle time variance amounted to 10,000 Adverse.

The actual fixed overheads incurred for the year amounted to ₹ 1,50,000 and the expenditure variance was ₹25,000 Favourable.

In the context of the above, the following needs to be determined:

- (i) The Actual output of Product Q produced during the year is:
- (a) 10,000 units
 - (b) 12,500 units
 - (c) 25,000 units
 - (d) 15,000 units

- (ii) The Material price and material cost variance are:
- (a) Price variance – 3,000 Adverse, Cost Variance – 5,000 Adverse
 - (b) Price variance – 3,000 Favourable, Cost Variance – 5,000 Favourable
 - (c) Price variance – 3,000 Favourable, Cost Variance – 8,000 Adverse
 - (d) Price variance – 5,000 Adverse, Cost Variance – 3,000 Favourable
- (iii) The Standard Hours, Net Actual hours and the idle time are:
- (a) Standard Hours – 27,500 Net Actual Hours – 28,000 hours Idle Time – 2,000 hours
 - (b) Standard Hours – 22,500 Net Actual Hours – 28,500 hours Idle Time – 1,500 hours
 - (c) Standard Hours – 24,000 Net Actual Hours – 29,000 hours Idle Time – 1,000 hours
 - (d) Standard Hours – 25,000 hours Net Actual Hours – 28,000 hours Idle Time – 2,000 hours
- (iv) Labour Efficiency variance and Labour rate variance are:
- (a) Labour Efficiency Variance – 30,000 Favourable Labour rate Variance – 25,000 Adverse
 - (b) Labour Efficiency Variance – 25,000 Favourable, Labour rate Variance – 30,000 Adverse
 - (c) Labour Efficiency Variance – 25,000 Adverse, Labour rate Variance – 30,000 Favourable
 - (d) Labour Efficiency Variance – 30,000 Adverse Labour rate Variance – 25,000 Favourable
- (v) Fixed Overhead volume variance is:
- (a) Fixed Overhead volume variance – 1,00,000 Favourable
 - (b) Fixed Overhead volume variance – 50,000 Adverse

- (c) Fixed Overhead volume variance – 1,00,000 Adverse
- (d) Fixed Overhead volume variance – 50,000 Favourable

Overheads: Absorption Costing Method

3. The accountant for Brilliant Tools Ltd applies overhead based on machine hours. The budgeted overhead and machine hours for the year are ₹ 1,30,000 and 8,000 hours, respectively. The actual overhead and machine hours incurred were ₹ 1,37,500 and 10,000 hours. The cost of goods sold and inventory data compiled for the year is as follows:

Direct Material ₹ 25,000

Cost of Goods Sold ₹ 2,25,000

Units: WIP 50,000 and Finished Goods 75,000

What is the amount of over/under absorbed overhead for the year?

- (a) Over absorbed by ₹ 25,000
- (b) Under absorbed by ₹ 25,000
- (c) Over a absorbed by ₹ 32,500
- (d) Under absorbed by ₹ 32,500

Process Costing

4. The following information is available in respect of Process I: Raw material purchased and introduced 10,000 units @ 5 per unit Raw Material received from store 4000 units @ 6 per unit Direct Labour 40,000 Overheads 28,000 Output of Process is 13,500 units, Normal wastage 5% of inputs Scrap value of wastage 4 per unit The value of Abnormal Gain is:

- (a) ₹ 2062.68
- (b) ₹ 2135.34
- (c) ₹ 2103.70
- (d) ₹ 2093.2

Service Costing

5. A hotel has 200 rooms (120 Deluxe rooms and 80 Premium rooms). The normal occupancy in summer is 80% and winter 60%. The period of summer and winter is taken as 8 months and 4 months respectively. Assume 30 days in each month. Room rent of Premium room will be double of Deluxe room. Hotel is expecting a profit of 20% on total revenue, total cost for the year is 2,66,11,200. Calculate the room rent to be charged for Premium room.
- (a) ₹ 450 per room day
 - (b) ₹ 900 per room day
 - (c) ₹ 380 per room day
 - (d) ₹ 760 per room day
6. ALC Ltd. is a insurance company. It launched a new term insurance policy Names as Protection Plus. The total cost for the policy during the year is ₹ 1,60,00,000. Total number of policies sold is 410 and total insured value of policies is ₹ 920 crore.
- What is the cost per rupee of insured value?
- (a) ₹ 0.0017
 - (b) ₹ 0.18
 - (c) ₹ 575
 - (d) ₹ 2.24

Budget and Budgetary Control

7. A business manufactures a single product and is preparing its production budget for the year ahead. It is estimated that 2,00,000 units of the product can be sold in the year and the opening inventory is currently 25,000 units. The inventory level is to be reduced by 40% by the end of the year. What is production budget in units?
- (a) 1,95,000 units
 - (b) 1,90,000 units
 - (c) 1,84,000 units

- (d) 1,75,000 units

Employee Cost

8. The labour turnover rates for the quarter ended 30th June, 2024 are computed as 14%, 8% and 6% under Flux method, Replacement method and Separation method respectively. If the number of workers replaced during 1st quarter of the financial year 2024-25 is 36, COMPUTE the following:
- The number of workers recruited and joined; and
 - The number of workers left and discharged.

Overheads: Absorption Costing Method

9. From the details furnished below you are required to compute a comprehensive machine-hour rate:

Original purchase price of the machine (subject to depreciation at 10% per annum on original cost)	₹ 12,96,000
Normal working hours for the month (The machine works for only 75% of normal capacity)	200 hours
Wages to Machine-man	₹ 800 per day (of 8 hours)
Wages to Helper (machine attendant)	₹ 500 per day (of 8 hours)
Power cost for the month for the time worked	₹ 1,30,000
Supervision charges apportioned for the machine centre for the month	₹ 18,000
Electricity & Lighting (fixed in nature) for the month	₹ 9,500
Repairs & maintenance (machine) including consumable stores per month	₹ 17,500
Insurance of Plant & Building (apportioned) for the year	₹ 18,000
Other general expense per annum	₹ 18,000

The workers are paid a fixed dearness allowance of ₹ 4,500 per month. Production bonus payable to workers in terms of an award is equal to 10% of basic wages and dearness allowance. Add 10% of the basic wage and dearness allowance against leave wages and holidays with pay to arrive at a comprehensive labour-wage for debit to production.

Activity Based Costing

10. SOFTHUG is a global brand created by Green-lush Ltd. The company manufactures three range of beauty soaps i.e. SOFTHUG- Gold, SOFTHUG- Pearl, and SOFTHUG- Diamond. The budgeted costs and production for the month of May, 2024 are as follows:

	SOFTHUG- Gold		SOFTHUG- Pearl		SOFTHUG- Diamond	
Production of soaps (Units)	4,000		3,000		2,000	
Resources per Unit:	Qty	Rate	Qty	Rate	Qty	Rate
- Essential Oils	60 ml	₹ 200/100 ml	55 ml	₹ 300/100 ml	65 ml	₹ 300/100 ml
- Cocoa Butter	20 g	₹ 200/100 g	20 g	₹ 200/100 g	20 g	₹ 200/100 g
- Filtered Water	30 ml	₹ 15/100 ml	30 ml	₹ 15/100 ml	30 ml	₹ 15/100 ml
- Chemicals	10 g	₹ 30/100 g	12 g	₹ 50/100 g	15 g	₹ 60/100 g
- Direct Labour	30 minutes	₹ 10/hour	40 minutes	₹ 10/hour	60 minutes	₹ 10 / hour

Green-lush Ltd. followed an Absorption Costing System and absorbed its production overheads, to its products using direct labour hour rate, which were budgeted at ₹ 1,98,000.

Now, Green-lush Ltd. is considering adopting an Activity Based Costing system. For this, additional information regarding budgeted overheads and their cost drivers is provided below:

Particulars	(₹)	Cost drivers
Forklifting cost	58,000	Weight of material lifted
Supervising cost	60,000	Direct labour hours
Utility cost	80,000	Number of Machine operations

The number of machine operators per unit of production are 5, 5, and 6 for SOFTHUG- Gold, SOFTHUG- Pearl, and SOFTHUG- Diamond respectively.

(Consider (i) Mass of 1 litre of Essential Oils and Filtered Water equivalent to 0.8 kg and 1 kg respectively (ii) Mass of output produced is equivalent to the mass of input materials taken together.)

You are required to:

- PREPARE a statement showing the unit costs and total costs of each product using the absorption costing method.
- PREPARE a statement showing the product costs of each product using the ABC approach.
- STATE what are the reasons for the different product costs under the two approaches?

Cost Sheet

11. From the following data of Appu Ltd., CALCULATE (i) Material Consumed; (ii) Prime Cost and (iii) Cost of production.

		Amount (₹)
(i)	Repair & maintenance paid for plant & machinery	9,80,500
(ii)	Insurance premium paid for inventories	26,000
(iii)	Insurance premium paid for plant & machinery	96,000
(iv)	Raw materials purchased	64,00,000
(v)	Opening stock of raw materials	2,88,000
(vi)	Closing stock of raw materials	4,46,000
(vii)	Wages paid	23,20,000
(viii)	Value of opening Work-in-process	4,06,000
(ix)	Value of closing Work-in-process	6,02,100

(x)	Quality control cost for the products in manufacturing process	86,000
(xi)	Research & development cost for improvement in production process	92,600
(xii)	Administrative cost for:	
	- Factory & production	9,00,000
	- Others	11,60,000
(xiii)	Amount realised by selling scrap generated during the manufacturing process	9,200
(xiv)	Packing cost necessary to preserve the goods for further processing	10,200
(xv)	Salary paid to Director (Technical)	8,90,000

Cost Accounting System

12. A manufacturing company disclosed a net loss of ₹ 3,47,000 as per their cost accounts for the year ended March 31,2024. The financial accounts however disclosed a net loss of ₹ 5,10,000 for the same period. The following information was revealed as a result of scrutiny of the figures of both the sets of accounts.

	(₹)
(i) Factory Overheads under-absorbed	40,000
(ii) Administration Overheads over-absorbed	60,000
(iii) Depreciation charged in Financial Accounts	3,25,000
(iv) Depreciation charged in Cost Accounts	2,75,000
(v) Interest on investments not included in Cost Accounts	96,000
(vi) Income-tax provided	54,000
(vii) Interest on loan funds in Financial Accounts	2,45,000
(viii) Transfer fees (credit in financial books)	24,000
(ix) Stores adjustment (credit in financial books)	14,000
(x) Dividend received	32,000

PREPARE a memorandum Reconciliation Account

Batch Costing

13. A jobbing factory has undertaken to supply 300 pieces of a component per month for the ensuing six months. Every month a batch order is opened against which materials and labour hours are booked at actual. Overheads are levied at a rate per labour hour. The selling price contracted for is ₹ 8 per piece. From the following data CALCULATE the cost and profit per piece of each batch order and overall position of the order for 1,800 pieces.

Month	Batch Output	Material cost	Direct wages	Direct labour
		(₹)	(₹)	hours
January	310	1150	120	240
February	300	1140	140	280
March	320	1180	150	280
April	280	1130	140	270
May	300	1200	150	300
June	320	1220	160	320

The other details are:

Month	Chargeable expenses	Direct labour
	(₹)	(Hours)
January	12,000	4,800
February	10,560	4,400
March	12,000	5,000
April	10,580	4,600
May	13,000	5,000
June	12,000	4,800

Process Costing

14. The following data are available in respect of Process-I for June 2024:
- (1) Opening stock of work in process: 600 units at a total cost of ₹ 4,20,000.

- (2) Degree of completion of opening work in process:
- | | |
|-----------|------|
| Material | 100% |
| Labour | 60% |
| Overheads | 60% |
- (3) Input of materials at a total cost of ₹ 55,20,000 for 9,200 units.
- (4) Direct wages incurred ₹ 18,60,000
- (5) Production overhead ₹ 8,63,000.
- (6) Units scrapped 200 units. The stage of completion of these units was:
- | | |
|-----------|------|
| Materials | 100% |
| Labour | 80% |
| Overheads | 80% |
- (7) Closing work in process; 700 units. The stage of completion of these units was:
- | | |
|-----------|------|
| Material | 100% |
| Labour | 70% |
| Overheads | 70% |
- (8) 8,900 units were completed and transferred to the next process.
- (9) Normal loss is 4% of the total input (opening stock plus units put in)
- (10) Scrap value is ₹ 60 per unit.

You are required to:

- COMPUTE equivalent production,
- CALCULATE the cost per equivalent unit for each element.
- CALCULATE the cost of abnormal loss (or gain), closing work in process and the units transferred to the next process using the FIFO method.

Joint Products & By-Products

15. Three products X, Y and Z alongwith a byproduct B are obtained again in a crude state which require further processing at a cost of ₹ 5 for X; ₹ 4 for Y; and ₹ 2.50 for Z per unit before sale. The byproduct is however saleable as such to a nearby factory. The selling prices for the three main products and byproduct, assuming they should yield a net margin of 25 percent of cost, are fixed at ₹ 13.75 ₹ 8.75 and ₹ 7.50 and ₹ 1.00 respectively – all per unit quantity sold.

During a period, the joint input cost including the material cost was ₹ 90,800 and the respective outputs were:

X	8,000 units
Y	6,000 units
Z	4,000 units
B	1,000 units

By product should be credited to the joint cost and only the net joint costs are to be allocated to the main products.

CALCULATE the joint cost per unit of each product and the margin available as a percentage on cost.

Service Costing

16. BK Infra Ltd. built and operates a 110 k.m. long highway on the basis of Built-Operate-Transfer (BOT) model for a period of 25 year. A traffic assessment has been carried out to estimate the traffic flow per day. The details are as below:

Sl. No.	Type of vehicle	Daily traffic volume
1.	Two wheelers	44,500
2.	Car and SUVs	3,450
3.	Bus and LCV	1,800
4.	Heavy commercial vehicles	816

The following is the estimated cost of the project:

Sl. No.	Activities	Amount (₹ in lakh)
1	Site clearance	170.70
2	Land development and filling work	9,080.35
3	Sub base and base courses	10,260.70
4	Bituminous work	35,070.80
5	Bridge, flyovers, underpasses, Pedestrian subway, footbridge, etc.	29,055.60
6	Drainage and protection work	9,040.50
7	Traffic sign, marking and road appurtenance	8,405.00
8	Maintenance, repairing and rehabilitation	12,429.60
9	Environmental management	982.00
	Total Project cost	114,495.25

An average cost of ₹ 1,120 lakh has to be incurred on administration and toll plaza operation.

On the basis of the vehicle specifications (i.e. weight, size, time saving etc.), the following weights has been assigned to the passing vehicles:

Sl. No.	Type of vehicle	
1.	Two wheelers	5%
2.	Car and SUVs	20%
3.	Bus and LCV	30%
4.	Heavy commercial vehicles	45%

Required:

- CALCULATE the total project cost per day of concession period.
- COMPUTE toll fee to be charged for per vehicle of each type, if the company wants to earn a profit of 15% on total cost.

[Note: Concession period is a period for which an infrastructure is allowed to operate and recovers its investment]

Marginal Costing

17. RS Ltd. manufactures and sells a single product X whose selling price is ₹ 100 per unit and the variable cost is ₹ 60 per unit.
- If the Fixed Costs for this year are ₹ 24,00,000 and the annual sales are at 60% margin of safety, CALCULATE the rate of net return on sales, assuming an income tax level of 40%
 - For the next year, it is proposed to add another product line Y whose selling price would be ₹ 150 per unit and the variable cost ₹ 100 per unit. The total fixed costs are estimated at ₹ 28,00,000. The sales mix of X : Y would be 5 : 3. COMPUTE the break-even sales in units for both the products.

Budget and Budgetary Control

18. Raja Ltd manufactures and sells a single product and has estimated sales revenue of ₹ 302.4 lakh during the year based on 20% profit on selling price. Each unit of product requires 6 kg of material A and 3 kg of material B and processing time of 4 hours in machine shop and 2 hours in assembly shop. Factory overheads are absorbed at a blanket rate of 20% of direct labour. Variable selling & distribution overheads are ₹ 60 per unit sold and fixed selling & distribution overheads are estimated to be ₹ 69,12,000.

The other relevant details are as under:

Purchase Price:	Material A	₹ 160 per kg
	Materials B	₹ 100 per kg

Labour Rate:	Machine Shop	₹ 140 per hour
	Assembly Shop	₹ 70 per hour

	Finished Stock	Material A	Material B
Opening Stock	2,500 units	7,500 kg	4,000 kg
Closing Stock	3,000 units	8,000 kg	5,500 kg

Required

- (i) CALCULATE number of units of product proposed to be sold and selling price per unit,
- (ii) PREPARE Production Budget in units and
- (iii) PREPARE Material Purchase Budget in units.

Miscellaneous

19. (a) DISCUSS the Net Realisable Value (NRV) method of apportioning joint costs to by-products.
- (b) DIFFERENTIATE between Service costing and Product costing.
- (c) DISCUSS the Controllable and un-controllable variances.
- (d) DISCUSS the Standard and Discretionary Cost Centres.


SUGGESTED ANSWERS/HINTS

1. (i) (d) Cash discount is received when credit amount is paid within the stipulated period of 30 days. The amount of cash discount to be received from the supplier is:

	Particulars	Amount (₹)
A.	Listed price	2,50,000
B.	Less: Trade Discount @10%	(25,000)
C.	Taxable value (A-B)	2,25,000
D.	Add: GST@18% (18% of C)	40,500
E.	Total amount payable to the supplier	2,65,500
F.	Cash discount @10% (10% of C)	(22,500)
G.	Net amount to be paid to the supplier (E-F)	2,43,000

(ii) (b)

Particulars	Units	(₹)
Listed Price of Materials	5,000	2,50,000
Less: Trade discount @ 10% on invoice price		(25,000)
		2,25,000
Add: GST @ 18% of ₹ 2,25,000		40,500
		2,65,500
Add: Toll Tax		5,000
Freight and Insurance		17,220
Commission and Brokerage Paid		10,000
Add: Cost of returnable containers: Amount deposited ₹ 30,000 Less: Amount refunded ₹ 20,000		10,000
		3,07,720
Add: Other Expenses @ 2% of Total Cost ($\frac{₹ 3,07,720}{98} \times 2$)		6,280
Total cost of material		3,14,000
Less: Shortage material due to normal reasons @ 20%	1,000	-
Total cost of material of good units	4,000	3,14,000
Cost per unit (₹ 3,14,000/4,000 units)		78.5

(iii) (b) Axe Traders is an unregistered supplier in the GST; thus, GST credit is not applicable for it. GST paid on the purchase of the material will be the part of the material cost.

(iv) (a) Please refer the solution above

(v) (c) Please refer the solution above

- 2. (i) (a)** 10,000 units
- Usage variance of Material Z = 2,000 F
- Usage Variance = $SQ \times SP - AQ \times SP$
- SP = ₹ 2
- AQ = 24,000 units
- $2 \times (SQ - 24,000) = 2,000$
- 2SQ = 50,000
- Therefore SQ = 25,000
- No of units of Input required per output = 2
- Yield of input = 80%
- $= (25000/2) \times 80\% = 10,000 \text{ units.}$
- (ii) (b)** Price variance – 3,000 Favourable,
Cost Variance – 5,000 Favourable
- Price variance = $AQ \times (SP - AP)$
- $24,000 \times (2 - 1.875) = 3,000 \text{ Favourable.}$
- Cost variance = $SQ \times SP - AQ \times AP$
- $= 50,000 - 45,000 = 5,000 \text{ Favourable.}$
- (iii) (d)** Standard Hours – 25,000 hours Net Actual Hours – 28,000 hours
Idle Time – 2,000 hours
- Actual output = 10,000 units
- Standard hours per unit = 2.5
- Therefore standard hours = $10,000 \times 2.5 = 25,000 \text{ hours.}$
- Idle time variance = $SR \times (\text{Net AH} - AH)$
- $5 \times (\text{Net AH} - 30,000) = 10,000 \text{ Adverse}$
- $5 \text{ Net AH} - 1,50,000 = -10,000$
- $5 \text{ Net AH} = 1,40,000$

Net AH = 28,000 hours

Idle time = 2,000 hours

(iv) (c) Labour Efficiency Variance – 25,000 Adverse,

Labour rate Variance – 30,000 Favourable

Efficiency Variance = $SR \times (SH - AH)$

= $5 \times (25,000 - 30,000)$

= 25,000 Adverse

Rate Variance = $AH \times (SR - AR)$

= $30,000 (5 - 4) [1,20,000/30,000]$

= 30,000 Favourable.

(v) (c) Fixed Overhead Volume variance – 1,00,000 Adverse

Overhead Volume variance = Actual Output $\times$ SR per unit – Budgeted FOH

Budgeted FOH = Actual FOH (+/-) Expenditure variance

$1,50,000 + 25,000 = 1,75,000$

$AO \times SR = 10,000 \times 7.5 = 75,000$

Therefore volume variance = $75,000 - 1,75,000$

= 1,00,000 Adverse.

3. (a) Overabsorbed by ₹ 25,000

Predetermined Overhead Rate = Budgeted Overhead / Budgeted hours i.e. $130,000 / 8,000 = ₹ 16.25$ per hour.

Hence, absorbed overhead = $10,000 \times 16.25 = ₹ 1,62,500$.

Since actual overhead incurred were ₹ 1,37,500

Hence the overhead were over absorbed by $1,62,500 - 1,37,500 = ₹ 25,000$.

4. (d) ₹ 2093.2

Process a/c

Particulars	Units	Amount	Particulars	units	Amount
Raw material	10,000	50,000	Normal loss	700	2,800
Stores	4,000	24,000	Units transferred	13,500	1,41,293.2
Direct Wages		40,000			
Production overheads		28,000			
Abnormal gain	200	2,093.2			
		1,44,093.2			1,44,093.2

$$\text{Cost per unit} = \frac{1,42,000 - 2,800}{14,000 - 700} = 10.466 \text{ per unit}$$

5. (b) ₹ 900 per room day

Total Revenue (2,66,11,200/80%) = 3,32,64,000

Calculation of Room Days:

	Deluxe	Premium
Summer	120 rooms x 80% x 30 days x 8 months = 23,040	80 rooms x 80% x 30 days x 8 months = 15,360
Winter	120 rooms x 60% x 30 days x 4 months = 8,640	80 rooms x 60% x 30 days x 4 months = 5,760
Total room days	31,680	21,120

Let's assume the room rent of Deluxe room be 'x'

Then rent of Premium room will be '2x'

Therefore: 31,680x + 42,240x = 3,32,64,000

X = 450

Rent of Premium room will be $450 \times 2 = ₹ 900$ per room day

6. (a) ₹ 0.0017

Cost per rupee of insured value

$$= \frac{\text{Total Cost}}{\text{Total Insured Value}}$$

$$= \frac{1.6 \text{ cr}}{920 \text{ cr}} = ₹ 0.0017$$

7. (b) 1,90,000 units

	Units
Sales budget	2,00,000
Add: Closing Inventory (25,000 x 0.6)	15,000
Less: Opening Inventory	(25,000)
Production Budget	1,90,000

8. Labour Turnover Rate (Replacement method) = $\frac{\text{No. of workers replaced}}{\text{Average No. of workers}}$

$$\text{Or, } \frac{8}{100} = \frac{36}{\text{Average No. of workers}}$$

$$\text{Or, Average No. of workers} = 450$$

$$\text{Labour Turnover Rate (Separation method)} = \frac{\text{No. of workers separated}}{\text{Average No. of workers}}$$

$$\text{Or, } \frac{6}{100} = \frac{\text{No. of workers separated}}{450}$$

$$\text{Or, No. of workers separated} = 27$$

$$\text{Labour Turnover Rate (Flux Method)} = \frac{\text{No. of Separations} + \text{No. of accession (Joinings)}}{\text{Average No. of workers}}$$

$$\text{Or, } \frac{14}{100} = \frac{27 + \text{No. of accessions (Joinings)}}{450}$$

$$\text{Or, } 100 (27 + \text{No. of Accessions}) = 6,300$$

$$\text{Or, No. of Accessions} = 36$$

$$(i) \quad \text{The No. of workers recruited and Joined} = 36$$

(ii) The No. of workers left and discharged = 27

9. Effective machine hours = 200 hours × 75% = 150 hours

Computation of Comprehensive Machine Hour Rate

	Per month (₹)	Per hour (₹)
Fixed cost		
Supervision charges	18,000.00	
Electricity and lighting	9,500.00	
Insurance of Plant and building (₹ 18,000 ÷ 12)	1,500.00	
Other General Expenses (₹ 18,000 ÷ 12)	1,500.00	
Depreciation (₹ 1,29,600 ÷ 12)	10,800.00	
	41,300.00	275.33
Direct Cost		
Repairs and maintenance	17,500.00	116.67
Power	1,30,000.00	866.67
Wages of machine man		196.00
Wages of Helper		136.00
Machine Hour rate (Comprehensive)		1,590.67

Wages per machine hour

	Machine man	Helper
Wages for 200 hours		
Machine-man (₹ 800 × 25)	₹ 20,000.00	---
Helper (₹ 500 × 25)	---	₹ 12,500.00
Dearness Allowance (DA)	₹ 4,500.00	₹ 4,500.00
	₹ 24,500.00	₹ 17,000.00

Production bonus (10% of Basic and DA)	2,450.00	1,700.00
Leave wages (10% of Basic and DA)	2,450.00	1,700.00
	29,400.00	20,400.00
Effective wage rate per machine hour	196.00	136.00

10. (i) Traditional Absorption Costing

	SOFTHUG-Gold	SOFTHUG - Pearl	SOFTHUG - Diamond	Total
(a) Production of soaps (Units)	4,000	3,000	2,000	9,000
(b) Direct labour (minutes)	30	40	60	-
(c) Direct labour hours (a × b)/60 minutes	2,000	2,000	2,000	6,000

Overhead rate per direct labour hour:

= Budgeted overheads ÷ Budgeted labour hours

= ₹ 1,98,000 ÷ 6,000 hours

= ₹ 33 per direct labour hour

Unit Costs:

	SOFTHUG-Gold (₹)	SOFTHUG-Pearl (₹)	SOFTHUG-Diamond (₹)
Direct Costs:			
- Direct Labour	5.00 $\left(\frac{10 \times 30}{60}\right)$	6.67 $\left(\frac{10 \times 40}{60}\right)$	10.00 $\left(\frac{10 \times 60}{60}\right)$
- Direct Material (Refer working note 1)	167.50	215.50	248.50

Production Overhead:	16.50 $\left(\frac{33 \times 30}{60}\right)$	22.00 $\left(\frac{33 \times 40}{60}\right)$	33.00 $\left(\frac{33 \times 60}{60}\right)$
Total unit costs	189.00	244.17	291.50
Number of units	4,000	3,000	2,000
Total costs	7,56,000	7,32,510	5,83,000

Working note -1
Calculation of Direct material cost

	SOFTHUG - Gold (₹)	SOFTHUG - Pearl (₹)	SOFTHUG - Diamond (₹)
Essential oils	120.00 $\left(\frac{200 \times 60}{100}\right)$	165.00 $\left(\frac{300 \times 55}{100}\right)$	195.00 $\left(\frac{300 \times 65}{100}\right)$
Cocoa Butter	40.00 $\left(\frac{200 \times 20}{100}\right)$	40.00 $\left(\frac{200 \times 20}{100}\right)$	40.00 $\left(\frac{200 \times 20}{100}\right)$
Filtered water	4.50 $\left(\frac{15 \times 30}{100}\right)$	4.50 $\left(\frac{15 \times 30}{100}\right)$	4.50 $\left(\frac{15 \times 30}{100}\right)$
Chemicals	3.00 $\left(\frac{30 \times 10}{100}\right)$	6.00 $\left(\frac{50 \times 12}{100}\right)$	9.00 $\left(\frac{60 \times 15}{100}\right)$
Total costs	167.50	215.50	248.50

(ii) Activity Based Costing

	SOFTHUG- Gold	SOFTHUG- Pearl	SOFTHUG- Diamond	Total
Quantity (units)	4,000	3,000	2,000	-
Weight per unit (grams)	108 $\{(60 \times 0.8) + 20 + 30 + 10\}$	106 $\{(55 \times 0.8) + 20 + 30 + 12\}$	117 $\{(65 \times 0.8) + 20 + 30 + 15\}$	-

Total weight (grams)	4,32,000	3,18,000	2,34,000	9,84,000
Direct labour (minutes)	30	40	60	-
Direct labour hours	2,000 $\left(\frac{4,000 \times 30}{60} \right)$	2,000 $\left(\frac{3,000 \times 40}{60} \right)$	2,000 $\left(\frac{2,000 \times 60}{60} \right)$	6,000
Machine operations per unit	5	5	6	-
Total operations	20,000	15,000	12,000	47,000

Forklifting rate per gram = ₹ 58,000 ÷ 9,84,000 grams = ₹ 0.06 per gram

Supervising rate per direct labour hour = ₹ 60,000 ÷ 6,000 hours = ₹ 10 per labour hour

Utilities rate per machine operations = ₹ 80,000 ÷ 47,000 machine operations
= ₹ 1.70 per machine operations

Unit Costs under ABC:

	SOFTHUG - Gold (₹)	SOFTHUG - Pearl (₹)	SOFTHUG - Diamond (₹)
Direct Costs:			
- Direct Labour	5.00	6.67	10.00
- Direct material	167.50	215.50	248.50
Production Overheads:			
Forklifting cost	6.48 (0.06 × 108)	6.36 (0.06 × 106)	7.02 (0.06 × 117)

Supervising cost	5.00 $\left(\frac{10 \times 30}{60}\right)$	6.67 $\left(\frac{10 \times 40}{60}\right)$	10.00 $\left(\frac{10 \times 60}{60}\right)$
Utilities	8.50 (1.70 × 5)	8.50 (1.70 × 5)	10.20 (1.70 × 6)
Total unit costs	192.48	243.70	285.72
Number of units	4,000	3,000	2,000
Total costs	7,69,920	7,31,100	5,71,440

- (iii) **Comments:** The difference in the total costs under the two systems is due to the differences in the overheads borne by each of the products. The Activity Based Costs appear to be more precise.

11. Calculation of Cost of Production of Appu Ltd.

Particulars	Amount (₹)
Raw materials purchased	64,00,000
Add: Opening stock	2,88,000
Less: Closing stock	(4,46,000)
Material consumed	62,42,000
Wages paid	23,20,000
Prime cost	85,62,000
Repair and maintenance cost of plant & machinery	9,80,500
Insurance premium paid for inventories	26,000
Insurance premium paid for plant & machinery	96,000
Quality control cost	86,000
Research & development cost	92,600
Administrative overheads related with factory and production	9,00,000
	1,07,43,100

Add: Opening value of W-I-P	4,06,000
Less: Closing value of W-I-P	(6,02,100)
	1,05,47,000
Less: Amount realised by selling scrap	(9,200)
Add: Primary packing cost	10,200
Cost of Production	1,05,48,000

Notes:

- (i) Other administrative overhead does not form part of cost of production.
- (ii) Salary paid to Director (Technical) is an administrative cost.

12. Memorandum Reconciliation Accounts

Dr.			Cr.
	(₹)		(₹)
To Net Loss as per Costing books	3,47,000	By Administration overheads over recovered in cost accounts	60,000
To Factory overheads under absorbed in Cost Accounts	40,000	By Interest on investment not included in Cost Accounts	96,000
To Depreciation under charged in Cost Accounts	50,000	By Transfer fees in financial books	24,000
To Income-Tax not provided in Cost Accounts	54,000	By Stores adjustment (Credit in financial books)	14,000
To Interest on Loan Funds in Financial Accounts	2,45,000	By Dividend received in financial books	32,000
		By Net loss as per financial books	5,10,000
	7,36,000		7,36,000

13. Statement of Cost and Profit per batch

Particulars	Jan.	Feb.	March	April	May	June	Total
Batch output (in units)	310	300	320	280	300	320	1,830
Sale value (₹)	2,480	2,400	2,560	2,240	2,400	2,560	14,640
Material cost (₹)	1,150	1,140	1,180	1,130	1,200	1,220	7,020
Direct wages (₹)	120	140	150	140	150	160	860
Chargeable expenses* (₹)	600	672	672	621	780	800	4,145
Total cost (₹)	1,870	1,952	2,002	1,891	2,130	2,180	12,025
Profit per batch (₹)	610	448	558	349	270	380	2,615
Total cost per unit (₹)	6.03	6.51	6.26	6.75	7.10	6.81	6.57
Profit per unit (₹)	1.97	1.49	1.74	1.25	0.90	1.19	1.43

Overall position of the order for 1,800 units

Sales value of 1,800 units @ ₹ 8 per unit ₹ 14,400

Total cost of 1,800 units @ ₹ 6.57 per unit ₹ 11,826

Profit ₹ 2,574

* $\frac{\text{Chargeable expenses}}{\text{Direct labour hour for the month}} \times \text{Direct labour hours for batch}$

14. (i) Statement of Equivalent Production (FIFO Method)

Input		Output		Equivalent Production					
				Materials		Labour		Production Overhead	
Details	Units	Details	Units	%	Units	%	Units	%	Units
Opening Stock	600	From opening stock	600	-	-	40	240	40	240

		- From fresh materials	8,300	100	8,300	100	8,300	100	8,300
		Closing W-I-P	700	100	700	70	490	70	490
Fresh inputs	9,200	Normal loss	392	-	-	-	-	-	-
			9,992		9,000		9,030		9,030
		Less: Abnormal Gain	(192)	100	(192)	100	(192)	100	(192)
	9,800		9,800		8,808		8,838		8,838

(ii) Statement of Cost per equivalent units

Elements	(₹)	Cost (₹)	Equivalent units (EU)	Cost per EU (₹)
Material Cost	55,20,000			
Less: Scrap realisation 392 units @ ₹ 60/- p.u.	(23,520)	54,96,480	8,808	624.03
Labour cost		18,60,000	8,838	210.45
Production OH Cost		8,63,000	8,838	97.65
Total Cost		82,19,480		932.13

(iii) Cost of Abnormal Gain – 192 Units

	(₹)	(₹)
Material cost of 192 units @ ₹ 624.03 p.u.	1,19,813.76	
Labour cost of 192 units @ ₹ 210.45 p.u.	40,406.40	
Production OH cost of 192 units @ ₹ 97.65 p.u.	18,748.80	1,78,968.96
Cost of closing WIP – 700 Units		
Material cost of 700 equivalent units @ ₹ 624.03 p.u.	4,36,821.00	
Labour cost of 490 equivalent units @ ₹ 210.45 p.u.	1,03,120.50	
Production OH cost of 490 equivalent @ ₹ 97.65 p.u.	47,848.50	5,87,790.00

Cost of 8,900 units transferred to next process	₹
(i) Cost of opening W-I-P Stock b/f – 600 units	4,20,000.00
(ii) Cost incurred on opening W-I-P stock	
Material cost	—
Labour cost 240 equivalent units @ ₹ 210.45 p.u.	50,508.00
Production OH cost 240 equivalent units @ ₹ 97.65 p.u.	
	<u>23,436.00</u>
	<u>4,93,944.00</u>
(iii) Cost of 8,300 completed units	
8,300 units @ ₹ 932.13 p.u.	<u>77,36,679.00</u>
Total cost [(i) + (ii) + (iii)]	<u>82,30,623.00</u>

15. Working Notes:
(i) Computation of Allocation Ratio for Joint Costs

	Products		
	X	Y	Z.
	₹	₹	₹
Selling Price	13.75	8.75	7.50
Less: anticipated margin@ 25% on cost of 20% on sales	2.75	1.75	1.50
Cost of sales	11.00	7.00	6.00
Less: post split off cost	5.00	4.00	2.50
Joint cost per unit	6.00	3.00	3.50
Output (units)	8,000	6,000	4,000
Total output cost	48,000	18,000	14,000
Allocation ratio for joint costs	24	9	7

(ii) Computation of net allocable joint costs

	₹	₹
Joint input cost including material cost		90,800
Less: Credit for realization from by-product B: Sales revenue (1,000 × Re. 1)	1,000	
Less: profit @ 25% on cost or 20% on sales	200	800
Net joint costs to be allocated		90,000

Determination of joint cost per unit of each product

Product	Net joint costs allocation ₹	Output (units) ₹	Joint cost per unit ₹
X	54,000 (Note: 1)	8,000	6.75
Y	20,250	6,000	3.38
Z	15,750	4,000	3.94
	90,000		

Profit margin available on each product as a percentage on cost

Product	Joint Cost ₹	Post spilt-off cost ₹	Total Cost ₹	Selling Price ₹	Margin ₹	Margin % on cost ₹
X	6.75	5.00	11.75	13.75	2.00	17.02
Y	3.38	4.00	7.38	8.75	1.37	18.56
Z	3.94	2.50	6.44	7.50	1.06	16.46

Note: 1

$$X = \frac{24}{40} \times 90,000 = 54,000$$

$$Y = \frac{9}{40} \times 90,000 = 20,250$$

$$Z = \frac{7}{40} \times 90,000 = \frac{15,750}{90,000}$$

16. (i) Calculation of total project cost per day of concession period:

Activities	Amount (₹ in lakh)
Site clearance	170.70
Land development and filling work	9,080.35
Sub base and base courses	10,260.70
Bituminous work	35,070.80
Bridge, flyovers, underpasses, Pedestrian subway, footbridge, etc.	29,055.60
Drainage and protection work	9,040.50
Traffic sign, marking and road appurtenance	8,405.00
Maintenance, repairing and rehabilitation	12,429.60
Environmental management	982.00
Total Project cost	114,495.25
Administration and toll plaza operation cost	1,120.00
Total Cost	115,615.25
Concession period in days (25 years × 365 days)	9,125
Cost per day of concession period (₹ in lakh)	12.67

(ii) Computation of toll fee:

$$\begin{aligned} \text{Cost to be recovered per day} &= \text{Cost per day of concession period} + 15\% \text{ profit on cost} \\ &= ₹ 12,67,000 + ₹ 1,90,050 \\ &= ₹ 14,57,050 \end{aligned}$$

$$\begin{aligned} \text{Cost per equivalent vehicle} &= \frac{₹ 14,57,050}{76,444 \text{ units (Refer workingnote)}} \\ &= ₹ 19.06 \text{ per equivalent vehicle} \end{aligned}$$

Vehicle type-wise toll fee:

Sl. No.	Type of vehicle	Equivalent cost [A]	Weight [B]	Toll fee per vehicle [A×B]
1.	Two wheelers	₹ 19.06	1	19.06
2.	Car and SUVs	₹ 19.06	4	76.24
3.	Bus and LCV	₹ 19.06	6	114.36
4.	Heavy commercial vehicles	₹ 19.06	9	171.54

Working Note:

The cost per day has to be recovered from the daily traffic. Each type of vehicle is to be converted into equivalent unit. Let's convert all vehicle types equivalent to Two-wheeler

Sl. No.	Type of vehicle	Daily traffic volume [A]	Weight	Ratio [B]	Equivalent Two-wheeler [A×B]
1.	Two wheelers	44,500	0.05	1	44,500
2.	Car and SUVs	3,450	0.20	4	13,800
3.	Bus and LCV	1,800	0.30	6	10,800
4.	Heavy commercial vehicles	816	0.45	9	7,344
	Total				76,444

17. (i) Contribution per unit = Selling price – Variable cost
 = ₹ 100 – ₹ 60
 = ₹ 40
- Break-even Point = $\frac{₹ 24,00,000}{₹ 40}$
 = 60,000 units
- Percentage Margin of Safety = $\frac{\text{Actual Sales} - \text{Break - even Sales}}{\text{Actual Sales}}$

$$\text{Or, } 60\% = \frac{\text{Actual Sales} - 60,000 \text{ units}}{\text{Actual Sales}}$$

$$\therefore \text{Actual Sales} = 1,50,000 \text{ units}$$

	(₹)
Sales Value (1,50,000 units × ₹ 100)	1,50,00,000
Less: Variable Cost (1,50,000 units × ₹ 60)	(90,00,000)
Contribution	60,00,000
Less: Fixed Cost	(24,00,000)
Profit	36,00,000
Less: Income Tax @ 40%	(14,40,000)
Net Return	21,60,000

$$\text{Rate of Net Return on Sales} = 14.40\% \left(\frac{₹ 21,60,000}{₹ 1,50,00,000} \times 100 \right)$$

(ii) Products

	X (₹)	Y (₹)
Selling Price <i>per unit</i>	100	150
Variable Cost <i>per unit</i>	60	100
Contribution <i>per unit</i>	40	50

Composite contribution will be as follows:

$$\text{Contribution per unit} = \left(\frac{40}{8} \times 5 \right) + \left(\frac{50}{8} \times 3 \right)$$

$$= 25 + 18.75 = ₹ 43.75$$

$$\text{Break-even Sale} = 64,000 \text{ units} \left(\frac{₹ 28,00,000}{₹ 43.75} \right)$$

Break-even Sales Mix:

$$X (64,000 \text{ units} \times 5/8) = 40,000 \text{ units}$$

$$Y (64,000 \text{ units} \times 3/8) = 24,000 \text{ units}$$

18. Workings
Statement Showing "Total Variable Cost for the year"

Particulars	Amount (₹)
Estimated Sales Revenue	3,02,40,000
Less: Desired Profit Margin on Sale @ 20%	60,48,000
Estimated Total Cost	2,41,92,000
Less: Fixed Selling and Distribution Overheads	69,12,000
Total Variable Cost	1,72,80,000

Statement Showing "Variable Cost per unit"

Particulars	Variable Cost p.u. (₹)
Direct Materials:	
A: 6 Kg. @ ₹ 160 per kg.	960
B: 3 Kg. @ ₹ 100 per kg.	300
Labour Cost:	
Machine Shop: 4 hrs @ ₹ 140 per hour	560
Assembly Shop: 2 hrs @ ₹ 70 per hour	140
Factory Overheads: 20% of (₹ 560 + ₹ 140)	140
Variable Selling & Distribution Expenses	60
Total Variable Cost per unit	2,160

(i) Calculation of number of units of product proposed to be sold and selling price per unit:

$$\begin{aligned}
 \text{Number of Units Sold} &= \text{Total Variable Cost} / \text{Variable Cost per unit} \\
 &= ₹ 1,72,80,000 / ₹ 2,160 \\
 &= 8,000 \text{ units} \\
 \text{Selling Price per unit} &= \text{Total Sales Value} / \text{Number of Units Sold} \\
 &= ₹ 3,02,40,000 / 8,000 \text{ units} \\
 &= ₹ 3,780
 \end{aligned}$$

(ii) Production Budget (units)

Particulars	Units
Budgeted Sales	8,000
Add: Closing Stock	3,000
Total Requirements	11,000
Less: Opening Stock	(2,500)
Required Production	8,500

(iii) Materials Purchase Budget (Kg.)

Particulars	Material A	Material B
Requirement for Production	51,000 (8,500 units × 6 Kg.)	25,500 (8,500 units × 3 Kg.)
Add: Desired Closing Stock	8,000	5,500
Total Requirements	59,000	31,000
Less: Opening Stock	(7,500)	(4,000)
Quantity to be purchased	51,500	27,000

- 19. (a) Net Realisable Value method:** The realisation on the disposal of the by-product may be deducted from the total cost of production so as to arrive at the cost of the main product. For example, the amount realised by the sale of molasses in a sugar factory goes to reduce the cost of sugar produced in the factory.

When the by-product requires some additional processing and expenses are incurred in making it saleable to the best advantage of the concern, the expenses so incurred should be deducted from the total value realised from the sale of the by-product and only the net realisations should be deducted from the total cost of production to arrive at the cost of production of the main product. Separate accounts should be maintained for collecting additional expenses incurred on:

- (i) further processing of the by-product, and

- (ii) selling, distribution and administration expenses attributable to the by-product.
- (b) Service costing differs from product costing (such as job or process costing) in the following ways due to some basic and peculiar nature.
 - (i) Unlike products, services are intangible and cannot be stored, hence, there is no inventory for the services.
 - (ii) Use of Composite cost units for cost measurement and to express the volume of outputs.
 - (iii) Unlike a product manufacturing, employee (labour) cost constitutes a major cost element than material cost.
 - (iv) Indirect costs like administration overheads are generally have a significant proportion in total cost of a service as unlike manufacturing sector, service sector heavily depends on support services and traceability of costs to a service may not economically feasible
- (c) **Controllable and un-controllable variances:** The purpose of the standard costing reports is to investigate the reasons for significant variances so as to identify the problems and take corrective action.

Variances are broadly of two types, namely, controllable and uncontrollable. Controllable variances are those which can be controlled by the departmental heads whereas uncontrollable variances are those which are beyond their control. Responsibility centres are answerable for all adverse variances which are controllable and are appreciated for favourable variances. Controllability is a subjective matter and varies from situation to situation. If the uncontrollable variances are of significant nature and are persistent, the standard may need revision.
- (d) (i) **Standards Cost Centres:** Cost Centre where output is measurable and input required for the output can be specified. Based on a well-established study, an estimate of standard units of input to produce a unit of output is set. The actual cost for inputs is compared with the standard cost. Any deviation

(variance) in cost is measured and analysed into controllable and uncontrollable cost. The manager of the cost centre is supposed to comply with the standard and held responsible for adverse cost variances. The input-output ratio for a standard cost centre is clearly identifiable.

- (ii) **Discretionary Cost Centre:** The cost centre whose output cannot be measured in financial terms, thus input-output ratio cannot be defined. The cost of input is compared with allocated budget for the activity. Example of discretionary cost centres are Research & Development department, Advertisement department where output of these department cannot be measured with certainty and co-related with cost incurred on inputs.



PAPER – 5: AUDITING AND ETHICS



QUESTIONS

PART – I: Multiple Choice Questions

Case Scenario

Pluto Limited is engaged in the manufacturing and distribution of furniture. After Covid, as the number of people working from home has gone up, this customisable range home office furniture has gained lot of importance. They were able to perform very well over the years and the same has been reflected in their financial statements.

During the year 2023-24, M/s Saha and Associates was reappointed as the auditor. The new engagement team has CA Saha (partner) and five article assistants. Given the large volume of transactions, the partner instructed the article assistants to review the financial statements and auditor's report from the previous year, 2022-23, to gain a thorough understanding.

While reviewing the summary page, one of the article Kabir noticed that few points were under discussion with the partner before finalising the audit:

The team verified the following points with respect to employee benefit expenses:

- The employee benefit expenses recorded in the books were actually incurred during the relevant period.
- The expenses in respect of all personnel have been accounted for.
- The expenses recognised during the period are pertaining to the current accounting period.

It was also noted that dividend to equity shareholders for the year 2022-23, was declared on 15.04.2023 and was recognised as liability in the year 2022-23.

Another article Krish noticed that debtors constitute a major component of the company's financial statements. As part of audit procedure, the auditor requested the client to obtain external confirmations from the parties. For this, a list of all the debtors were obtained and a random sampling was performed by the auditor. The client directly obtained the selected debtor confirmations from the customers on time.

Krish also observed addition in the assets amounting ₹ 50 lakhs during the year supported by sufficient audit evidence. However, upon scrutiny, it was found that some of the invoices were not in the name of the company. Further, Cash and cash equivalents were classified as balances with banks, cheques and drafts on hand, cash on hand and earmarked balances with banks (e.g. unpaid dividend)

Based on above, answer the following questions: -

1. Which among the following assertions are discussed with respect to the employee benefit expenses?

- (i) Measurement
- (ii) Occurrence
- (iii) Cutoff
- (iv) Completeness

Choose the correct combination from below:

- (a) (i), (ii) and (iv)
- (b) (ii), (iii) and (iv)
- (c) (i), (iii) and (iv)
- (d) (i), (ii) and (iii)

2. Whether disclosure of dividend under the liability head during the year 2022-23 is appropriate?
- (a) Yes. It should be recognised in the same year irrespective of the year of declaration as the amount of dividend belongs to year 2022-23.
 - (b) No. The amount should be recognised equally between two financial years.
 - (c) No. The amount should not be recognised as a liability. It should be disclosed in the notes to accounts.
 - (d) No. The amount should neither be recognised as liability nor disclosure is required in the financial statements.
3. Krish pointed out that that the method followed to obtain debtor confirmation in the previous year was not in accordance with SA 505. Therefore, M/s Saha & Associated should reperform the same in the correct manner.
- (a) As per SA 505, confirmation should be directly obtained by the auditor. Further, for all significant account balances as on the Balance sheet date confirmations should necessarily be collected and for the smaller outstanding balances, random sampling could be performed.
 - (b) It is ok to obtain confirmation through clients as they are in constant contact with their customers. Also, many customers may not respond to auditor's external confirmation request mail. Random selection can be done for all debtors irrespective of the amount in accordance with SA 505.
 - (c) As per SA 505, confirmation should be directly obtained by the auditor. Whereas random selection can be done for all debtors irrespective of the amount.
 - (d) Either auditor or client can obtain confirmation based on time availability. However, it is necessary to obtain confirmation for all significant account balances as on the Balance sheet date and for the remaining random sampling could be performed in accordance with SA 505.

4. Which assertion has been affected in the case of fixed assets?
- (a) Existence
 - (b) Rights and obligation
 - (c) Completeness
 - (d) Measurement
5. Cash and cash equivalents were not properly classified by the client. Which of the following is incorrect disclosure of the same?
- (a) Balances with banks
 - (b) Cheques and drafts on hand
 - (c) Cash on hand
 - (d) Earmarked balances with banks (e.g.: unpaid dividend)

General MCQs

6. CA Chatur provides accounting and book keeping services to Shiv Ganga Limited. He is also offered an audit of the accounts to the Shiv Ganga Limited. Which threat to independence may be involved in accepting such an engagement?
- (a) Self-interest threats
 - (b) Self-review threats
 - (c) Familiarity threats
 - (d) Advocacy threats
7. CA Sargun is conducting statutory audit of branch of KBC Bank. During the course of audit, it is noticed that one borrower has availed a housing loan and a car loan from the branch. Housing loan EMIs are overdue for 120 days as on date of Balance sheet. Car loan EMIs are overdue for 60 days as on date of Balance sheet. Which of the following statements is appropriate?
- (a) Housing Loan should be classified as "Non-Performing Asset" in accordance with RBI norms on asset classification. However, Car loan should be classified as Standard asset.

- (b) Car Loan should be classified as "Non-Performing Asset" in accordance with RBI norms on asset classification. However, Housing Loan should be classified as Standard asset.
 - (c) Both Housing loan as well as car loan should be classified as "Non-Performing Assets" in accordance with RBI norms on asset classification.
 - (d) Both Housing as well as car loans should be classified as Standard assets.
8. Sagar Finance Corporation Limited is a government company. The audit of the company is conducted by statutory auditors appointed by Comptroller and Auditor General of India (C&AG). Who is empowered to conduct "supplementary audit" of Sagar Finance Corporation Ltd.?
- (a) Central Government
 - (b) C&AG
 - (c) Another independent auditor appointed by C&AG
 - (d) Another independent auditor appointed by Ministry of Corporate Affairs

PART II – Descriptive Questions**Chapter 1 – Nature, Objective and Scope of Audit**

9. Kriti, a CA student, has joined articles in a reputed audit firm. She considers audit engagement to be an "assurance engagement". Her understanding is that audit engagement is the only kind of assurance engagement in which practitioner gives a written assurance report in appropriate form. However, her friend Somaya, does not agree with her. She is of the view that assurance engagements are not restricted to audit alone. Besides, Somaya also thinks that assurance engagements can also relate to matters other than historical financial information. Whose view appears to be correct? State with reasons.

Chapter 2 – Audit Strategy, Audit Planning and Audit Programme

10. CA N, engagement partner of LPS & Associates, is planning for audit of a large company. As part of preliminary engagement activities being performed in this regard, he wants to ensure that compliance with

independence requirements is adhered. How shall he form a conclusion that audit firm complies with independence requirements?

Chapter 3 – Risk Assessment and Internal Control

11. Companies are increasingly using data analytics for managing their operations. Auditors can also obtain good results by using data analytics. What are the tools and techniques used by auditors in applying the principles of Data Analytics and in which areas data analytics can be used by auditors?

Chapter 4 – Audit Evidence

12. Quality Products Limited is engaged in manufacturing of bicycles. As part of manufacturing activities, it sends raw material to some business entities and procures finished components from them. As on 31st March 2024 inventories lying with such business entities are material. Being statutory auditor of Quality Products Limited, how you will obtain sufficient appropriate audit evidence regarding existence and condition of inventories lying with such business entities?

Chapter 5 – Audit of Items of Financial Statements

13. The financial statements of XYZ Limited reflects long-term borrowings from the banks, financial institutions, leasing, and hire purchase companies. Additionally, the company has issued debentures to its 1000 members to raise funds in accordance with the provisions of the Companies Act, 2013. The money raised by issuing debentures is also reflected in long-term borrowings. As the statutory auditor of XYZ Limited, CA X wants to verify that all borrowings on the balance sheet represent valid claims by banks or other third parties. Suggest a few audit procedures in this regard.

Chapter 6 – Audit Documentation

14. An important factor in determining the form, content and extent of audit documentation of significant matters is the extent of professional judgment exercised in performing the work and evaluating the results. Explain stating clearly the examples of significant matters.

Chapter 7 – Completion and Review

15. CA Manoj is the statutory auditor of a company. He has completed the audit and issued the audit report dated June 30, 2024. Before signing the audit report, written representations dated June 29, 2024, were obtained from the management of the company. Discuss the significance of the date of written representations. Also state whether the written representations obtained on June 29, 2024 is appropriate or not.
16. While conducting audit of BYN Limited, CA Y notices that company has lost one of its key markets along with important customers. Additionally, several highly successful competitors have emerged, impacting business of the company. Despite the existence of material uncertainty, CA Y finds the use of going concern basis of accounting appropriate for preparation of financial statements. The company has also disclosed material uncertainty in notes to accounts adequately. How should he deal with the matter in auditor's report?

Chapter 8 – Audit Report

17. Discuss the reporting requirements as per CARO, 2020, regarding:
- (i) Statutory Dues
 - (ii) Transactions with Related Parties

Chapter 9 – Special Features of Audit of Different Type of Entities

18. Section 77 of Multi-State Cooperative Societies Act, 2002 states that the Central Government may at any time by order direct that a special audit of the Multi-State Co-Operative Society's accounts for such period or periods as may be specified in the order, shall be conducted. Further, it may by the same or a different order appoint either a Chartered Accountant or the Multi-State Co-Operative Society's auditor himself to conduct special audit. Under which circumstances such power can be exercised by the Central Government?

Chapter 10 – Audit of Banks

19. Compute the Drawing Power for Cash Credit A/c of Kirpa Limited for the month of March 2024 with following information:

(Amount in ₹)

Stock	60,000
Debtors	55,000
(Including Debtor of ₹ 15,000 for an invoice dated 20.10.2023)	
Sundry creditors	10,000
Sanctioned Limit	48,000

Margin on stock is 30% and on debtors is 40%.

Note: Debtors older than 3 months are ineligible for calculation of DP.

Chapter 11 – Ethics and Terms of Audit Engagements

20. ABC & Associates, an audit firm, has been approached by a prospective company client that has been in business for about 10 years to conduct an audit of its financial statements. Before accepting the audit engagement, the firm wants to assess the integrity of prospective client. With regard to the assessment of integrity, which matters should be considered by the audit firm?



SUGGESTED ANSWERS/HINTS

PART – I: Answers to Multiple Choice Questions

MCQ No.	Answer
1.	(b)
2.	(c)
3.	(a)
4.	(b)
5.	(d)
6.	(b)
7.	(c)
8.	(b)

PART – II: Answers to Descriptive Questions

9. Audit engagement is an assurance engagement. However, assurance engagements are not restricted to auditing alone. There are other assurance engagements too like review engagements, engagements providing assurance involving prospective financial information, engagements providing assurance on internal controls in an entity. Assurance engagements provide assurance to users. The difference is of degree. Whereas an audit provides reasonable assurance which is a high level of assurance, review provides lower level of assurance as compared to audit.

There are also assurance engagements which relate to matters other than historical financial information like providing assurance on matters involving prospective financial information and providing assurance on internal controls in an entity. In assurance reports involving prospective financial information, the practitioner obtains sufficient appropriate evidence to the effect that management's assumptions on which the prospective financial information is based are not unreasonable, the prospective financial information is properly prepared on the basis of the assumptions, and it is properly presented, and all material assumptions are adequately disclosed.

In the given case, Kriti is of the view that audit engagement is the only kind of assurance engagement in which practitioner gives a written assurance report in appropriate form whereas is of the view that assurance engagements are not restricted to audit alone. Thus, view of Somaya is correct.

10. The engagement partner, CA N shall form a conclusion on compliance with independence requirements that apply to the audit engagement. In doing so, CA N shall: -
- (i) Obtain relevant information from the firm to identify and evaluate circumstances and relationships that create threats to independence
 - (ii) Evaluate information on identified breaches, if any, of the firm's independence policies and procedures to determine whether they create a threat to independence for the audit engagement and

- (iii) Take appropriate action to eliminate such threats or reduce them to an acceptable level by applying safeguards, or, if considered appropriate, to withdraw from the audit engagement, where withdrawal is permitted by law or regulation. The engagement partner shall promptly report to the firm any inability to resolve the matter for appropriate action.
- 11.** The tools and techniques that auditors use in applying principles of data analytics are known as Computer Assisted Auditing Techniques (CAATs). Data analytics can be used in testing of electronic records and data residing in IT systems using spreadsheets and specialised audit tools viz., IDEA and ACL to perform the following: -
- Check completeness of data and population that is used in either test of controls or substantive audit tests.
 - Selection of audit samples – random sampling, systematic sampling.
 - Re-computation of balances – reconstruction of trial balance from transaction data.
 - Re-performance of mathematical calculations–depreciation, bank interest calculation.
 - Analysis of journal entries
 - Fraud investigation.
 - Evaluating impact of control deficiencies.
- 12.** In accordance with requirements of SA 501, "Audit Evidence- Specific Considerations for Selected Items", when inventory under the custody and control of a third party is material to the financial statements, the auditor shall obtain sufficient appropriate audit evidence regarding the existence and condition of that inventory by performing one or both of the following:
- (a) Request confirmation from the third party as to the quantities and condition of inventory held on behalf of the entity.
 - (b) Perform inspection or other audit procedures appropriate in the circumstances.

Other audit procedure may include –

- Inspecting documentation regarding inventory held by third parties.
- Requesting confirmation from other parties when inventory has been pledged as collateral.
- Attending, or arranging for another auditor to attend, the third party's physical counting of inventory, if practicable.
- Obtaining another auditor's report, or a service auditor's report, on the adequacy of the third party's internal control for ensuring that inventory is properly counted and adequately safeguarded.

13. The audit procedures required to be undertaken by CA X while auditing borrowings is as follows:

- Review board minutes for approval of new lending agreements. During review, ensure that new loan agreements or bond issuances were authorized. Ensure that significant debt commitments are approved by the board of directors.
- Check the details of loans recorded (interest rate, nature and repayment terms) to the loan agreement. Verify that borrowing limits, if any, imposed by the agreements are not exceeded.
- Roll out and obtain independent balance confirmations (SA 505) in respect of all the borrowings from the lender (banks/ financial institutions etc.).
- Verify the details of leases and hire purchase creditors recorded to underlying contracts/agreements.
- In case of Debentures, examine trust deed for terms and dates of redemption, borrowing restrictions and compliance with covenants.
- When debt is retired, ensure that a discharge is received on assets securing the debt.
- Obtain Written Representation that all the liabilities which have been recorded represent a valid claim by the lenders.

- 14.** As per SA 230, "Audit documentation" judging the significance of a matter requires an objective analysis of the facts and circumstances.

Examples of significant matters include:

- Matters that give rise to significant risks.
- Results of audit procedures indicating (a) that the financial statements could be materially misstated, or (b) a need to revise the auditor's previous assessment of the risks of material misstatement and the auditor's responses to those risks.
- Circumstances that cause the auditor significant difficulty in applying necessary audit procedures.
- Findings that could result in a modification to the audit opinion or the inclusion of an Emphasis of Matter Paragraph in the auditor's report.

An important factor in determining the form, content and extent of audit documentation of significant matters is the extent of professional judgement exercised in performing the work and evaluating the results.

Documentation of the professional judgements made, where significant, explains the auditor's conclusions and reinforces the quality of the judgement.

Such matters are of particular interest to those responsible for reviewing audit documentation, including those carrying out subsequent audits, when reviewing matters of continuing significance (for example, when performing a retrospective review of accounting estimates).

- 15.** As per SA 580, "Written Representations" the date of the written representations shall be as near as practicable to, but not after, the date of the auditor's report on the financial statements.

Since written representations are necessary audit evidence, the auditor's opinion cannot be expressed, and the auditor's report cannot be dated before the date of the written representations. Furthermore, because the auditor must consider events occurring up to the date of the auditor's report that might require adjustment to or disclosure in the financial statements, the written representations are dated as near as practicable

to, but not after, the date of the auditor's report on the financial statements.

In the given situation, CA Manoj obtained written representations dated June 29, 2024, from the management of the company before signing the audit report on June 30, 2024.

From the above it can be concluded that written representations have been obtained appropriately.

- 16.** As per SA 570, "Going concern", If adequate disclosure about the material uncertainty is made in the financial statements, the auditor shall express an unmodified opinion and the auditor's report shall include a separate section under the heading "Material Uncertainty Related to Going Concern" to: -

- (a) Draw attention to the note in the financial statements that discloses such matters.
- (b) State that these events or conditions indicate that a material uncertainty exists that may cast significant doubt on the entity's ability to continue as a going concern and that the auditor's opinion is not modified in respect of the matter.

- 17. (i) Reporting under CARO, 2020 regarding statutory Dues [Para 3(vii)]**

- (a) whether the company is regular in depositing undisputed statutory dues including Goods and Services Tax, provident fund, employees' state insurance, income tax, sales-tax, service tax, duty of customs, duty of excise, value added tax, cess and any other statutory dues to the appropriate authorities and if not, the extent of the arrears of outstanding statutory dues as on the last day of the financial year concerned for a period of more than six months from the date they became payable, shall be indicated;
- (b) where statutory dues referred to in sub-clause (a) have not been deposited on account of any dispute, then the amounts involved and the forum where dispute is pending

shall be mentioned (a mere representation to the concerned Department shall not be treated as a dispute);

(ii) Reporting under CARO,2020 regarding transactions with related parties [Para 3(xiii)]

Whether all transactions with the related parties are in compliance with sections 177 and 188 of Companies Act where applicable and the details have been disclosed in the financial statements, etc., as required by the applicable accounting standards.

18. The Central government has power to order special audit of Multi-State Cooperative Society's Accounts where it is of the opinion -

- (a) that the affairs of any multi-state co-operative society are not being managed in accordance with self-help and mutual deed and co-operative principles or prudent commercial practices or with sound business principles or ;
- (b) that any multi-state co-operative society is being managed in a manner likely to cause serious injury or damage to the interests of the trade industry or business to which it pertains or ;
- (c) that the financial position of any multi-state co-operative society is such as to endanger its solvency.

19. Computation of Drawing Power (DP) for CC A/c of Kirpa Ltd.

Particulars of current assets		Amount (₹)	DP Amt (₹)
(A) Stocks:			
Stocks at realizable value		60,000	
Less: Unpaid stocks:			
- Sundry creditors	10,000	<u>10,000</u>	
Paid for stocks		50,000	
Margin @ 30%		<u>15,000</u>	35,000
(B) Debtors:			
Total Debtors		55,000	
Less: Ineligible debtors		<u>15,000</u>	

Eligible debtors		40000	
Margin @ 40%		<u>16,000</u>	<u>16,000</u>
Total Drawing Power			51,000

The sanctioned limit given in the question is ₹ 48,000 whereas drawing power as per the above working is ₹ 51,000. So, drawing power would be restricted to sanctioned limit i.e., ₹ 48,000/-.

20. With regard to the integrity of a client, matters that ABC & Associates should consider include, for example:

- The identity and business reputation of the client's principal owners, key management, related parties and those charged with its governance.
- The nature of the client's operations, including its business practices.
- Information concerning the attitude of the client's principal owners, key management and those charged with its governance towards such matters as aggressive interpretation of accounting standards and the internal control environment.
- Whether the client is aggressively concerned with maintaining the firm's fees as low as possible.
- Indications of an inappropriate limitation in the scope of work.
- Indications that the client might be involved in money laundering or other criminal activities.



PAPER – 6: FINANCIAL MANAGEMENT AND STRATEGIC MANAGEMENT

6A: FINANCIAL MANAGEMENT



QUESTIONS

Division A: Case Scenario

Investment Decision

1. Mr. Ronak, a doctor by profession, has his own private hospital at Goa having specialization in cardiac treatments. However, now-a-days, Goa not only being a place for the tourists, but is also a place for business delegates, cultural people, politicians, students and other classes of people. Gradually, Goa is opening new windows for businesses and getting recognition as an important tourist and leisure hub in South West India.

There are a number of hotels and resorts at Goa. However, the need still exists for more hotel services, in particular with the excellent service, and because of the large number of visitors from all over the country and all walks of life always favour Goa state for their recreation.

Mr. Ronak although being a doctor by profession is contemplating to establish a five-star hotel at Goa. The hotel will consist of 5 floors. The hotel will include 40 normal rooms and 8 deluxe suites, as well as a restaurant and couple of conference rooms with a small wedding hall on the ground floor. Following are the estimated occupancy rate including

fare composition in the Table 1. Being a five-star hotel, breakfast would be complementary but lunch and dinner are on a-la-carte basis.

Table 1: Hotel accommodation, estimated occupancy rate and fare.

Types of Facility	Numbers	Occupancy Rate	Average Rent Per Room Per Day	Growth Rate in Rent
Normal Room	40	33% or 120 Days	₹ 8000	12%
Deluxe Suites	8	33% or 120 Days	₹ 25,000	9%
Conference with Wedding Hall	2	40 days	₹ 3,00,000	9%
Restaurant	1	All days	₹ 27,000 sales per day	8%

For the sake of simplicity in calculation, growth rate to be applied only once after completion of 10 years.

The estimated cost of land will be ₹ 250 million and the construction cost will be ₹ 100 million. The estimated salvage value at the end of 15th year will be 25% of the cost of construction. The cost of furniture will be of ₹ 1,50,000 for each normal room and ₹ 3,80,000 for each deluxe suite. The cost of the furniture for the conference room with wedding hall will be ₹ 7,00,000 each and for restaurant it will be 10,00,000. In addition, the hotel will require 4 elevators at different locations and will be costing around ₹ 35,00,000 each. The cost of buying and installing electronic appliances like TV sets, Air conditioners, Fridge etc. will be around ₹ 35 million. Elevators would be depreciated at a rate of 5% p.a. Electronic appliances will have a salvage value of 15% of its acquisition cost at the end of 15 years.

The hotel will be built by renowned builder named '**Harihar Infrastructure**'. The builder estimated that building will survive for 15 years. The required furniture will be supplied by the local reputed furniture company named **Veru Furnishings Ltd.** They ensured that furniture will go for 10 years very smoothly. At the end of tenth year, new furniture for normal rooms and deluxe suites will be bought and old

furniture for the same will be sold by the hotel owner. The owner of the hotel estimates that he would be able to purchase the required furniture at 15% higher price than the previous purchase price. The salvage values of the furniture at the end of tenth year will be 5% of their purchase prices with no book value remaining. Furniture at restaurant, conference and wedding hall will not require any major changes as such except for minor renovation which will cost ₹ 20,00,000 in total at the end of 12th year. Any scrap generated on account of such renovation will be sold at ₹ 1,75,000.

In order to boost the tourism industry at Goa, the state govt will be granting subsidy of 15% on the initial capex incurred, it will be paid at the time of cost incurred and additional subsidy of 10% on annual revenue expenses for the first 3 years of operation, but will be credited directly in the bank account only at the end of 5th year and the same shall be non-taxable.

The total annual recurring expenses will be ₹ 1,80,00,000/-. It includes salaries to managers, staff and employees, utilities expenses, house keeping and security services' contract, AMC for electronic appliances, restaurant supplies and materials, other miscellaneous expenses, etc.

After the end of 10 years, annual recurring expenses will increase at a rate of 10% which is to be applied once. Furthermore, the hotel authority is determined to provide the best and professional hotel services to the clients by offering training to the employees. They decided to spend ₹ 5,00,000 per year for the purpose of training of the employees.

The hotel project will be entitled to enjoy tax holiday for the first five years after which the corporate tax rate of 25% will also be applied for the hotel. The Cost of equity for the company is 12% and the estimated hurdle rate by considering the structure of capital of the proposed hotel is fixed at 15%.

(Depreciation to be taken on SLM basis and assume 360 days in a year. Ignore depreciation on furniture used in restaurant, conference and wedding hall)

Based on above, please answer to the following MCQs.

- (i) The amount of net initial investment required is:
 - (a) ₹ 41.044 Crores
 - (b) ₹ 34.887 Crores
 - (c) ₹ 6.156 Crores
 - (d) ₹ 40.74 Crores
- (ii) NPV of the project is:
 - (a) ₹ 7.0532 Cr
 - (b) ₹ 8.4029 Cr
 - (c) ₹ 8.4935 Cr
 - (d) ₹ 2.4700 Cr
- (iii) Pay Back period of the project to recover the initial investment is:
 - (a) 5.12 years
 - (b) 12.02 years
 - (c) 11.80 years
 - (d) 4.46 years
- (iv) Estimated Recurring accounting profit/(loss) for first three years are:
 - (a) ₹ 7.0928 Cr p.a
 - (b) ₹ 6.9078 Cr p.a
 - (c) ₹ 6.9937 Cr p.a
 - (d) ₹ 9.6120 Cr p.a
- (v) IRR of the project is:
 - (a) 16.25%
 - (b) 19.39%
 - (c) 15%
 - (d) 12%

Ratio Analysis

2. KT Ltd.'s opening stock was ₹ 2,50,000 and the closing stock was ₹ 3,75,000. Sales during the year were ₹ 13,00,000 and the gross profit ratio was 25% on sales. Average accounts payable are ₹ 80,000. Creditors Turnover Ratio =?
- (a) 13.33
(b) 14.33
(c) 14.44
(d) 13.75

Leverage

3. A firm has sales of ₹ 75,00,000, variable cost of ₹ 42,00,000 and fixed cost of ₹ 6,00,000.
- It has a debt of ₹ 45,00,000 at 9% and equity of ₹ 55,00,000. Does it have favourable financial leverage?
- (a) ROI is less than interest on loan funds and hence it has no favourable financial leverage.
(b) ROI is equal to interest on loan funds and hence it has favourable financial leverage.
(c) ROI is greater than interest on loan funds and hence it has favourable financial leverage.
(d) ROI is greater than interest on loan funds and hence it has unfavourable financial leverage.

Division B: Descriptive Questions
Ratio Analysis

4. Following are the data in respect of LP enterprises for the year ended 31st March, 2024:
- | | | |
|---------------------------------|---|------|
| Debt to Total assets ratio | : | 0.40 |
| Long-term debts to equity ratio | : | 30% |
| Gross profit margin on sales | : | 20% |

Accounts receivables period	:	36 days
Quick ratio	:	0.9
Inventory holding period	:	60 days
Cost of goods sold	:	₹ 64,00,000

Liabilities	₹	Assets	₹
Equity Share Capital	20,00,000	Fixed assets	
Reserves & surplus		Inventories	
Long-term debts		Accounts receivable	
Accounts payable		Cash	
Total	50,00,000	Total	

Required:

COMPLETE the Balance Sheet of LP enterprises as on 31st March, 2024.
All calculations should be in nearest Rupee. Assume 360 days in a year.

Cost of Capital

5. BS Ltd. has the following capital structure at book-value as on 31st March, 2024:

Particulars	(₹)
Equity share capital (10,00,000 shares)	3,00,00,000
11.5% Preference shares	60,00,000
10% Debentures	1,00,00,000
	4,60,00,000

The equity shares of the company are sold for ₹ 300. It is expected that the company will pay next year a dividend of ₹ 15 per equity share, which is expected to grow by 5% p.a. forever. Assume a 35% corporate tax rate.

Required:

- (i) COMPUTE weighted average cost of capital (WACC) of the company based on the existing capital structure.
- (ii) COMPUTE the new WACC, if the company raises an additional ₹ 50 lakhs debt by issuing 10 years 12% debentures but the yield on debentures of similar maturity and risk class is 13%; flotation cost is 2%. Face value of the debenture is ₹100. This would result in increasing the expected equity dividend to ₹ 20 and leave the growth rate unchanged, but the price of equity share will fall to ₹ 250 per share.

Capital Structure

6. Company XYZ is unlevered and has a cost of equity of 20 percent and a total market value of ₹ 10,00,00,000. Company ABC is identical to XYZ in all respects except that it uses debt finance in its capital structure with a market value of ₹ 4,00,00,000 and a cost of 10 percent. FIND the market value of equity, weighted average cost of capital and cost of equity of ABC if the tax advantage of debt is 25 percent.
7. The following data relate to two companies belonging to the same risk class:

Particulars	A Ltd.	B Ltd.
Expected Net Operating Income	₹ 18,00,000	₹ 18,00,000
12% Debt	₹ 54,00,000	-
Equity Capitalization Rate	-	18

Required:

- (a) DETERMINE the total market value, Equity capitalization rate and weighted average cost of capital for each company assuming no taxes as per M.M. Approach.
- (b) DETERMINE the total market value, Equity capitalization rate and weighted average cost of capital for each company assuming 40% taxes as per M.M. Approach.

Leverage

8. Following data of PC Ltd. under Situations 1, 2 and 3 and Financial Plan A and B is given:

Installed Capacity (units)	3,600
Actual Production and Sales (units)	2,400
Selling price per unit (₹)	30
Variable cost per unit (₹)	20
Fixed Costs (₹):	
Situation 1	3,000
Situation 2	6,000
Situation 3	9,000

Capital Structure:

Particulars	Financial Plan	
	A	B
Equity	₹ 15,000	₹ 22,500
Debt	₹ 15,000	₹ 7,500
Cost of Debt	12%	12%

Required:

- CALCULATE the operating leverage and financial leverage.
- FIND out the combinations of operating and financial leverage which give the highest value and the least value.

Dividend Decision

9. The following information is taken from Gamma Ltd.

Net Profit for the year	₹ 30,00,000
12% Preference share capital	₹ 1,00,00,000
Equity share capital (Share of ₹ 10 each)	₹ 60,00,000
Internal rate of return on investment	22%
Cost of Equity Capital	18%

Retention Ratio 75%

CALCULATE the market price of the share using:

- (1) Gordon's Model
- (2) Walter's Model

Working Capital

10. TMT Limited is commencing a new project for manufacture of electric toys. The following cost information has been ascertained for annual production of 60,000 units at full capacity:

		Amount per unit (₹)
Raw materials		20
Direct labour		15
Manufacturing overheads:		
	₹	
Variable	15	
Fixed	<u>10</u>	25
Selling and Distribution overheads:		
	₹	
Variable	3	
Fixed	<u>1</u>	<u>4</u>
Total cost		64
Profit		<u>16</u>
Selling price		<u>80</u>

In the first year of operations expected production and sales are 40,000 units and 35,000 units respectively. To assess the need of working capital, the following additional information is available:

- (i) Stock of Raw materials 3 months consumption.
- (ii) Credit allowable for debtors 1½ months.
- (iii) Credit allowable by creditors 4 months.

- (iv) Lag in payment of wages 1 month.
- (v) Lag in payment of overheads $\frac{1}{2}$ month.
- (vi) Cash in hand and Bank is expected to be ₹ 60,000.
- (vii) Provision for contingencies is required @ 10% of working capital requirement including that provision.

You are required to PREPARE a projected statement of working capital requirement for the first year of operations. Debtors are taken at cost.

11. Banu Limited is considering relaxing its present credit policy and is in the process of evaluating two proposed policies. Currently, the firm has annual credit sales of ₹ 225 lakhs and accounts receivable turnover ratio of 5 times a year. The current level of loss due to bad debts is ₹ 7,50,000. The firm is required to give a return of 20% on the investment in new accounts receivables. Policy option II requires a manager to manage the receivables with salary of ₹ 50,000 per month. The company's variable costs are 60% of the selling price. Given the following information, which is a better option?

(Amount in lakhs)

	Present Policy	Policy Option I	Policy Option II
Annual credit sales (₹)	225	275	350
Accounts receivable turnover ratio	5	4	3
Bad debt losses (₹)	7.5	22.5	47.5

Miscellaneous

12. (a) EXPLAIN the advantages and disadvantages of both profit maximization and wealth maximization goals
- (b) EXPLAIN in brief the features of various types of preference shares.



SUGGESTED ANSWERS/HINTS

Division A: Case Scenario

1. (i) (b) ₹ 34.887 Crores

Amount Initial Investment required:

(A) Cost of Land & Construction Cost = 250 + 100 = 350 million i.e 35,00,00,000

(B) Furniture Cost

Normal Rooms = 40 x 1,50,000 = 60,00,000

Suite rooms = 8 x 3,80,000 = 30,40,000

Conference and wedding halls = 2 x 7,00,000
= 14,00,000

Restaurant = 10,00,000

(C) Elevators = 4 x 35,00,000 = 1,40,00,000

(D) Electronic Appliances = 3,50,00,000

Gross Investment Required = ₹ 41,04,40,000

Less: 15% Govt Subsidy on Capex = ₹ (6,15,66,000)

Net Initial Investment to be incurred by Hotel
= ₹ 34,88,74,000

(ii) (a) ₹ 7.0532 Cr

PV of Cash Inflow = ₹ 42.2317 Cr

As per WN - 2

(-) PV of Cash Outflow = ₹ 35.1785 Cr

As per WN - 1

NPV = 7.0532 CR

Note: Discounting Rate would be the hurdle rate and not cost of equity as hurdle rate means the overall cost of capital

WN 1 - Calculation of PV of Cash Outflows

		(₹)	DF @ 15%	PV (₹)
Year 1	Initial Net Investment	4,88,74,000	1.0000	34,88,74,000
At the end of 10th Year				
Year 10	Purchase of new furniture (At 15% higher price)			
	Normal Rooms	69,00,000	0.2472	17,05,680
	Suite Rooms	34,96,000	0.2472	8,64,211
At the end of 12th Year				
Year 12	Renovation at restaurant, conference and wedding halls (Net)	18,25,000	0.1869	3,41,093
				35,17,84,983

WN – 2: Calculation of PV of Cash Inflows

	Year 1 to 5 (₹)	Year 6 to 10 (₹)	Year 11 to 15 (₹)
Sales			(Apply growth rate here)
Normal Rooms	3,84,00,000	3,84,00,000	4,30,08,000
Suites	2,40,00,000	2,40,00,000	2,61,60,000
Conf & Hall	2,40,00,000	2,40,00,000	2,61,60,000
Restaurant	97,20,000	97,20,000	1,04,97,600
Total Sales (A)	9,61,20,000	9,61,20,000	10,58,25,600
Less:			
Annual Recurring Exp (Excl Depreciation)	1,80,00,000	1,80,00,000	1,98,00,000
Training Exp	5,00,000	5,00,000	5,00,000

(+) PV of Other Cash Inflows

Annual Revenue Exp

In year 15 - Salvage of building and electronic appliance = $3,02,50,000 \times 0.1229$
= ₹ 37,17,725

Therefore, **Total PV of Cash Inflows** = ₹ 42,23,17,089

Total Net Initial Investment Incurred = ₹ 34,88,74,000

Cumulative of Total Cash Inflows (not discounted cash inflows)

Year	Total Cash Inflows (₹)	Cumulative of Cash Inflows (₹)
1.	7,76,20,000	7,76,20,000
2.	7,76,20,000	15,52,40,000
3.	7,76,20,000	23,28,60,000
4.	7,76,20,000	31,04,80,000
5.	8,31,70,000 (Govt Subsidy of 55,50,000 added here)	39,36,50,000
6.	6,03,50,533	45,40,00,533

From the above table, it can be seen that the initial net investment incurred is getting recovered after 4th year but before the end of 5th year i.e. somewhere between 4th & 5th Year.

$$\text{Payback period} = 4 + \frac{3,83,94,000}{8,31,70,000}$$

$$= 4 + 0.46 = 4.46 \text{ years}$$

(iv) (a) ₹ 7.0928 Cr p.a.

Accounting profit = NPAT + Govt Subsidy on Revenue Expense

$$= 6,90,77,867 + (1,85,00,000 \times 10\%)$$

$$= 6,90,77,867 + 18,50,000$$

$$= 7,09,27,867 \text{ per annum}$$

(v) (b) 19.39%

DF @ 15%

$$\text{NPV} = ₹ 7.0532 \text{ Cr}$$

DF @ 20%

$$\text{NPV} = 34,10,68,926.17 - 35,07,57,719.00$$

$$= (₹ 0.9688 \text{ Cr})$$

$$\begin{aligned} \text{IRR} &= \text{Lower Rate \%} + \\ &\quad \left\{ \frac{\text{NPV @ Lower rate \%} \times (\text{Higher Rate \%} - \text{Lower Rate \%})}{\text{NPV @ Lower rate \%} - \text{NPV @ Higher Rate \%}} \right\} \\ \text{IRR} &= 15 + (7.0532 \times 5 / 8.0220) \\ &= 15 + 4.396 \\ &= 19.396\% \end{aligned}$$

2. (d) 13.75

$$\begin{aligned} \text{Creditors Turnover Ratio} &= \text{Purchases} / \text{Average Accounts Payable} \\ \text{Cost of Goods Sold} &= \text{Opening Stock} + \text{Purchases} - \text{Closing Stock} \\ \text{Purchases} &= \text{Cost of Goods Sold} + \text{Closing Stock} - \text{Opening Stock} \\ \text{Purchases} &= ₹ 9,75,000 + ₹ 3,75,000 - ₹ 2,50,000 \\ \text{Purchases} &= ₹ 11,00,000 \\ \text{Average Accounts Payable} &= ₹ 80,000 \\ \text{Creditors Turnover Ratio} &= \text{Purchases} / \text{Average Accounts Payable} \\ \text{Creditors Turnover Ratio} &= ₹ 11,00,000 / ₹ 80,000 \\ \text{Creditors Turnover Ratio} &= 13.75 \end{aligned}$$

Therefore, the Creditors Turnover Ratio is 13.75.

3. (c) ROI is greater than interest on loan funds and hence it has favourable financial leverage.

$$\text{EBIT} = 75,00,000 - 42,00,000 - 6,00,000 = 27,00,000,$$

$$\text{ROI} = 27,00,000 / (45,00,000 + 55,00,000) = 27\%,$$

Rate of Interest lower than Return on investment.

Therefore, there is favourable leverage.

Division B: Descriptive Questions

4. Working Notes:

$$(1) \quad \text{Total liability} = \text{Total Assets} = ₹ 50,00,000$$

- Debt to Total Asset Ratio = 0.40
- $$\frac{\text{Debt}}{\text{Total Assets}} = 0.40$$
- $$\frac{\text{Debt}}{50,00,000} = 0.40$$
- So, **Debt** = **20,00,000**
- (2) Total Liabilities = ₹ 50,00,000
- Equity share Capital + Reserves + Debt = ₹ 50,00,000
- So, Reserves = ₹ 50,00,000 - ₹ 20,00,000 - ₹ 20,00,000
- So, Reserves & Surplus** = **₹ 10,00,000**
- (3) $\frac{\text{Long term Debt}}{\text{Equity Shareholders' Fund}} = 30\%$
- $$\frac{\text{Long term Debt}}{(20,00,000 + 10,00,000)} = 30\%$$
- Long Term Debt** = **₹ 9,00,000**
- (4) So, Accounts Payable = ₹ 20,00,000 – ₹ 9,00,000
- Accounts Payable** = **₹ 11,00,000**
- (5) Gross Profit to sales = 20%
- Cost of Goods Sold = 80% of Sales = ₹ 64,00,000
- Sales** = $\frac{100}{80} \times 64,00,000$ = **80,00,000**
- (6) Inventory Turnover = $\frac{360}{60}$
- $$\frac{\text{COGS}}{\text{Closing inventory}} = \frac{360}{60}$$
- $$\frac{64,00,000}{\text{Closing inventory}} = \frac{360}{60}$$
- Closing inventory** = **10,66,667**
- (7) Accounts Receivable period = 36 days

$$\frac{\text{Accounts Receivable}}{\text{Credit sales}} \times 360 = 36$$

$$\begin{aligned} \text{Accounts Receivable} &= \frac{36}{360} \times \text{credit sales} \\ &= \frac{36}{360} \times 80,00,000 \text{ (assumed all sales are on credit)} \end{aligned}$$

$$\text{Accounts Receivable} = \text{₹ } 8,00,000$$

$$(8) \text{ Quick Ratio} = 0.9$$

$$\frac{\text{Quick Assets}}{\text{Current liabilities}} = 0.9$$

$$\frac{\text{Cash + Debtors}}{11,00,000} = 0.9$$

$$\text{Cash} + 8,00,000 = \text{₹ } 9,90,000$$

$$\text{Cash} = \text{₹ } 1,90,000$$

$$(9) \text{ Fixed Assets} = \text{Total Assets} - \text{Current Assets} = 50,00,000 - (10,66,667 + 8,00,000 + 1,90,000)$$

$$(10) = \text{29,43,333}$$

Balance Sheet of LP enterprises as on 31st March 2024

Liabilities	(₹)	Assets	(₹)
Share Capital	20,00,000	Fixed Assets	29,43,333
Reserved surplus	10,00,000	Current Assets:	
Long Term Debt	9,00,000	Inventory	10,66,667
Accounts Payable	11,00,000	Accounts Receivables	8,00,000
		Cash	1,90,000
Total	50,00,000	Total	50,00,000

(*Note: Equity shareholders' fund represent equity in 'Long term debts to equity ratio'. The question can be solved assuming only share capital as 'equity')

5. (i) **Computation of Weighted Average Cost of Capital based on existing capital structure**

Source of Capital	Existing Capital structure (₹)	Weights (a)	After tax cost of capital (%) (b)	WACC (%) (a) × (b)
Equity share capital (W.N.1)	3,00,00,000	0.652	10.00	6.52
11.5% Preference share capital	60,00,000	0.130	11.50	1.50
10% Debentures (W.N.2)	1,00,00,000	0.218	6.50	1.42
Total	4,60,00,000	1.000		9.44

Working Notes:

1. Cost of Equity Capital:

$$\begin{aligned}
 K_e &= \frac{\text{Expected dividend}(D_1)}{\text{Current Market Price}(P_0)} + \text{Growth}(g) \\
 &= \frac{₹ 15}{₹ 300} + 0.05 \\
 &= 10\%
 \end{aligned}$$

2. Cost of 10% Debentures

$$\begin{aligned}
 K_d &= \frac{\text{Interest}(1-t)}{\text{Net proceeds}} \\
 &= \frac{₹ 10,00,000(1-0.35)}{₹ 1,00,00,000} \\
 &= 0.065 \text{ or } 6.5\%
 \end{aligned}$$

(ii) **Computation of Weighted Average Cost of Capital based on new capital structure**

Source of Capital	New Capital structure (₹)	Weights (a)	After tax cost of capital (%) (b)	WACC (%) (a) x (b)
Equity share capital (W.N.3)	3,00,00,000	0.588	13.00	7.64
11.5% Preference share capital	60,00,000	0.118	11.50	1.36
10% Debentures (W.N.2)	1,00,00,000	0.196	6.50	1.27
12% Debentures (W.N.4)	50,00,000	0.098	9.21	0.90
Total	5,10,00,000	1.000		11.17

Working Notes:

3. Cost of Equity Capital:

$$K_e = \frac{\text{₹ } 20}{\text{₹ } 250} + 0.05$$

$$= 13\%$$

4. Cost of 12% Debentures

$$K_d = \frac{I(1-t) + \left(\frac{RV - NP}{n} \right)}{\frac{RV + NP}{2}}$$

$$= \frac{\text{₹ } 12(1-0.35) + \left(\frac{\text{₹ } 100 - \text{₹ } 90.31^*}{10 \text{ years}} \right)}{\frac{\text{₹ } 100 + \text{₹ } 90.31^*}{2}}$$

$$= \frac{\text{₹ } 8.769}{\text{₹ } 95.155} = 0.0921$$

*Since yield on similar type of debentures is 13 per cent, the company would be required to offer debentures at discount.

Market price of debentures (approximation method)

$$= ₹ 12 \div 0.13 = ₹ 92.31$$

Sale proceeds from debentures = ₹ 92.31 – ₹ 2 (i.e., floatation cost) = ₹ 90.31

6. Computation of Market Value of Equity of Company ABC

Total market value of Company ABC

$$V_{ABC} = V_{XYZ} + Bt \dots \dots \dots (i)$$

Where,

V_{ABC} = Market value of leveraged company.

V_{XYZ} = Market value of unleveraged company.

B = Market value of debt.

t = Tax rate.

Now, given

$$V_{xyz} = ₹ 10,00,00,000$$

$$B = ₹ 4,00,00,000$$

$$t = 25\%$$

By substituting values in equation (i) above, we have

$$\begin{aligned} V_{ABC} &= ₹ 10,00,00,000 + ₹ 4,00,00,000 \times 0.25\% \\ &= ₹ 11,00,00,000 \end{aligned}$$

The Market Value of Equity (s) of Company ABC,

$$\begin{aligned} &= ₹ 11,00,00,000 - ₹ 4,00,00,000 \\ &= ₹ 7,00,00,000 \end{aligned}$$

Weighted Average Cost of Capital of Company ABC

$$WACC_{ABC} = WACC_{XYZ} [1 - Bt/V_{ABC}]$$

$$= 20\% \left[1 - \frac{4,00,00,000}{11,00,00,000} \times 0.25 \right]$$

$$= 18.18\%$$

Where,

$WACC_{ABC}$ is the weighted average cost of capital of the levered company ABC

$WACC_{XYZ}$ is the weighted average cost of capital of the unlevered company XYZ.

Cost of Equity of company ABC

$$R_{Eabc} = R_{Exyz} + [(1 - t)B/E(R_{Exyz} - R_B)]$$

$$20\% + [(1 - .25)4,00,00,000/7,00,00,000(.20 - .10)]$$

$$24.28\% \text{ approx.}$$

Where,

R_{EABC} is the cost of equity in the levered Company ABC.

R_{Exyz} is the cost of equity in the unlevered Company XYZ.

E is the market value of equity.

B is the market value of debt.

R_B is the cost of debt

7. (a) Assuming no tax as per MM Approach.

Calculation of Value of Firms 'A Ltd.' and 'B Ltd' according to MM Hypothesis

Market Value of 'B Ltd' [Unlevered(u)]

$$\text{Total Value of Unlevered Firm } (V_u) = [NOI/k_e] = 18,00,000/.18$$

$$= ₹ 1,00,00,000$$

$$K_e \text{ of Unlevered Firm (given)} = 0.18$$

$$K_o \text{ of Unlevered Firm (Same as above} = k_e \text{ as there is no debt)}$$

$$= 0.18$$

Market Value of 'A Ltd' [Levered Firm (l)]

$$\begin{aligned}\text{Total Value of Levered Firm } (V_L) &= V_u + (\text{Debt} \times \text{Nil}) = ₹ 1,00,00,000 \\ &+ (54,00,000 \times \text{nil}) \\ &= ₹ 1,00,00,000\end{aligned}$$

**Computation of Equity Capitalization Rate and
Weighted Average Cost of Capital (WACC)**

	Particulars	A Ltd.	B Ltd.
A	Net Operating Income (NOI)	18,00,000	18,00,000
B	Less: Interest on Debt (I)	6,48,000	-
C	Earnings of Equity Shareholders (NI)	11,52,000	18,00,000
D	Overall Capitalization Rate (k_o)	0.18	0.18
E	Total Value of Firm ($V = \text{NOI}/k_o$)	1,00,00,000	1,00,00,000
F	Less: Market Value of Debt	54,00,000	-
G	Market Value of Equity (S)	46,00,000	1,00,00,000
H	Equity Capitalization Rate [$k_e = \text{NI} / S$]	0.2504	0.18
I	Weighted Average Cost of Capital [WACC] (k_o)	0.18	0.18

Assuming 40% taxes as per MM Approach

Calculation of Value of Firms 'A Ltd.' and 'B Ltd' according to MM Hypothesis Approach

Market Value of 'B Ltd' [Unlevered(u)]

$$\text{Total Value of unlevered Firm } (V_u) = [\text{NOI}(1 - t)/k_e] = 18,00,000 (1 - 0.40) / 0.18$$

$$= ₹ 60,00,000$$

$$K_e \text{ of unlevered Firm (given)} = 0.18$$

$$K_o \text{ of unlevered Firm (Same as above} = k_e \text{ as there is no debt)} = 0.18$$

Market Value of 'A Ltd' [Levered Firm (I)]

$$\begin{aligned}
 \text{Total Value of Levered Firm (V}_L) &= V_u + (\text{Debt} \times \text{Tax}) \\
 &= ₹ 60,00,000 + (54,00,000 \times 0.40) \\
 &= ₹ 81,60,000
 \end{aligned}$$

Computation of Weighted Average Cost of Capital (WACC) of 'B Ltd.'

$$= 18\% \text{ (i.e. } K_e = K_o)$$

Computation of Equity Capitalization Rate and Weighted Average Cost of Capital (WACC) of A Ltd

Particulars	A Ltd.
Net Operating Income (NOI)	18,00,000
Less: Interest on Debt (I)	6,48,000
Earnings Before Tax (EBT)	11,52,000
Less: Tax @ 40%	4,60,800
Earnings for equity shareholders (NI)	6,91,200
Total Value of Firm (V) as calculated above	81,60,000
Less: Market Value of Debt	54,00,000
Market Value of Equity (S)	27,60,000
Equity Capitalization Rate [$k_e = NI/S$]	.2504
Weighted Average Cost of Capital (k_o)*	13.23

*Computation of WACC A Ltd

Component of Capital	Amount	Weight	Cost of Capital	WACC
Equity	27,60,000	0.338	0.2504	0.0846
Debt	54,00,000	0.662	0.072*	0.0477
Total	81,60,000			0.1323

$$*K_d = 12\% (1 - 0.4) = 12\% \times 0.6 = 7.2\%$$

$$\text{WACC} = 13.23\%$$

8. (i) Operating Leverage

	Situation 1	Situation 2	Situation 3
	(₹)	(₹)	(₹)
Sales (S)			
2,400 units @ ₹ 30 per unit	72,000	72,000	72,000
Less: Variable Cost (VC) @ ₹ 20 per unit	48,000	48,000	48,000
Contribution (C)	24,000	24,000	24,000
Less: Fixed Cost (FC)	3,000	6,000	9,000
EBIT	21,000	18,000	15,000
Operating Leverage $= \frac{C}{EBIT}$	$\frac{₹ 24,000}{₹ 21,000}$ = 1.14	$\frac{₹ 24,000}{₹ 18,000}$ = 1.33	$\frac{₹ 24,000}{₹ 15,000}$ = 1.60

Financial Leverage

	Financial Plan	
	A (₹)	B (₹)
Situation 1		
EBIT	21,000	21,000
Less: Interest on debt (₹ 15,000 x 12%); (₹ 7,500 x 12%)	1,800	900
EBT	19,200	20,100
Financial Leverage = $\frac{EBIT}{EBT}$	$\frac{₹ 21,000}{₹ 19,200} = 1.09$	$\frac{₹ 21,000}{₹ 20,100} = 1.04$
Situation 2		
EBIT	18,000	18,000
Less: Interest on debt	1,800	900
EBT	16,200	17,100
Financial Leverage = $\frac{EBIT}{EBT}$	$\frac{₹ 18,000}{₹ 16,200} = 1.11$	$\frac{₹ 18,000}{₹ 17,100} = 1.05$

Situation 3		
EBIT	15,000	15,000
Less: Interest on debt	1,800	900
EBT	13,200	14,100
Financial Leverage = $\frac{EBIT}{EBT}$	$\frac{₹ 15,000}{₹ 13,200} = 1.14$	$\frac{₹ 15,000}{₹ 14,100} = 1.06$

(ii) Combined Leverages

$$CL = OL \times FL$$

		Financial Plan	
		A (₹)	B (₹)
(a)	Situation 1	$1.14 \times 1.09 = 1.24$	$1.14 \times 1.04 = 1.19$
(b)	Situation 2	$1.33 \times 1.11 = 1.48$	$1.33 \times 1.05 = 1.40$
(c)	Situation 3	$1.60 \times 1.14 = 1.82$	$1.60 \times 1.06 = 1.70$

The above calculations suggest that the highest value is in Situation 3 financed by Financial Plan A and the lowest value is in the Situation 1 financed by Financial Plan B.

9. Market price per share by-
(1) Gordon's Model:

$$\text{Present market price per share } (P_o)^* = \frac{D_o(1+g)}{K_e - g}$$

OR

$$\text{Present market price per share } (P_o) = \frac{D_1}{K_e - g}$$

Where,

P_o = Present market price per share.

g = Growth rate (br) = $0.75 \times 0.22 = 0.165$

b = Retention ratio (i.e., % of earnings retained)

r = Internal rate of return (IRR)

E = Earnings per share

$$P_o = \frac{0.75(1+0.165)}{0.18-0.165} = \frac{0.874}{0.015} = ₹ 58.27 \text{ approx.}$$

(2) Walter's Model:

$$P = \frac{D + \frac{r}{K_e}(E-D)}{K_e}$$

$$= \frac{0.75 + \frac{0.22}{0.18}(3-0.75)}{0.18} = ₹ 19.44$$

Workings:

1. Calculation of Earnings per share

Particulars	Amount (₹)
Net Profit for the year	30,00,000
Less: Preference dividend (12% of ₹ 1,00,00,000)	(12,00,000)
Earnings for equity shareholders	18,00,000
No. of equity shares (₹ 60,00,000/₹ 10)	6,00,000
Therefore, Earnings per share ($\frac{\text{Earning for equity shareholders}}{\text{No. of equity shares}}$)	₹ 18,00,000/6,00,000 = ₹ 3.00

2. Calculation of Dividend per share (D₀)

Particulars	
Earnings per share	₹ 3
Retention Ratio (b)	75%
Dividend pay-out ratio (1-b)	25%
Dividend per share (Earnings per share x Dividend pay-out ratio)	₹ 3 x 0.25 = ₹ 0.75

10. Statement Showing Cost and Sales for the First Year

Annual Production Capacity	60,000 units
Production	40,000 units
Sales	35,000 units

Particulars	₹
Sales Revenue (₹ 80 × 35,000)	28,00,000
Cost of Production:	
Materials @ ₹ 20 per unit	8,00,000
Direct Labour @ ₹ 15 per unit	6,00,000
Manufacturing Overheads	
Variable @ ₹ 15 per unit	6,00,000
Fixed (based on production capacity 60,000 units × ₹ 10)	<u>6,00,000</u>
Cost of Production	26,00,000
Less: Closing Stock (40,000 – 35,000 = 5,000 units)	
$\left(₹ \frac{26,00,000}{40,000} \times 5,000 \text{ units} \right)$	<u>3,25,000</u>
Cost of Goods Sold	22,75,000
Add: Selling & Distribution Overheads	
Variable @ ₹ 3 × 35,000 units = 1,05,000	
Fixed (Re. 1 × 60,000 units) = 60,000	<u>1,65,000</u>
Cost of Sales	<u>24,40,000</u>
Profit	<u>3,60,000</u>

Statement Showing Working Capital Requirement

A.	Current Assets	₹
	Stock of Raw Materials (₹ 8,00,000 × 3/12)	2,00,000
	Stock of Finished Goods	3,25,000
	Debtors at Cost (₹ 24,40,000 × 3/24)	3,05,000

	Cash and Bank	<u>60,000</u>
	Total (A)	<u>8,90,000</u>
B.	Current Liabilities	
	Creditors for Materials (₹ 10,00,000 × 4/12)	3,33,333
	Creditors for Expenses (₹ 13,65,000 × 1/24)	56,875
	Outstanding Wages (₹ 6,00,000 × 1/12)	<u>50,000</u>
	Total (B)	<u>4,40,208</u>
	Working Capital Requirement before Contingencies (A – B)	4,49,792
	Add: Provision for Contingencies (₹ 4,49,792 × 1/9)	<u>49,977</u>
	Estimated Working Capital Requirement	<u>4,99,769</u>

Workings Notes:

Purchase of Raw Material during the first year	₹
Raw Material consumed during the year	8,00,000
Add: Closing Stock of Raw Materials (3 months consumption)	<u>2,00,000</u>
	10,00,000
Less: Opening Stock of Raw Material	<u>Nil</u>
Purchases during the year	<u>10,00,000</u>

11. Statement showing Evaluation of Credit Policies (Amount in lakhs)

	Particulars	Present Policy (₹)	Proposed Policy I (₹)	Proposed Policy II (₹)
A	Expected Profit:			
	(a) Credit Sales	225.00	275.00	350.00
	(b) Total Cost other than Bad Debts:			
	Variable Costs other than manager salary	135.00	165.00	210.00

	(c) Salary of Manager	-	-	6
	(d) Bad Debts	7.50	22.50	47.50
	(e) Expected Profit [(a)-(b)-(c)-(d)]	82.50	87.50	86.50
B	Opportunity Cost of Investment in Receivables*	5.40	8.25	14.40
C	Net Benefits [A-B]	77.10	79.25	72.10

Recommendation: The Proposed Policy I should be adopted since the net benefits under this policy is higher than those under other policies.

Working Note:

***Calculation of Opportunity Cost of Average Investments**

$$\text{Opportunity Cost} = \text{Total Cost} \times \frac{\text{Collection Period}}{12} \times \frac{\text{Rate of Return}}{100}$$

$$\text{Present Policy} = ₹ 135 \text{ lakhs} \times 2.4/12 \times 20\% = ₹ 5.40 \text{ lakhs}$$

$$\text{Proposed Policy I} = ₹ 165 \text{ lakhs} \times 3/12 \times 20\% = ₹ 8.25 \text{ lakhs}$$

$$\text{Proposed Policy II} = ₹ 216 \text{ lakhs} \times 4/12 \times 20\% = ₹ 14.40 \text{ lakhs}$$

12. (a) The table below highlights some of the advantages and disadvantages of both profit maximization and wealth maximization goals:

Goal	Objective	Advantages	Disadvantages
Profit Maximization	Large amount of profits	(i) Easy to calculate profits (ii) Easy to determine the link between financial decisions and profits.	(i) Emphasizes the short term gains (ii) Ignores risk or uncertainty (iii) Ignores the timing of returns (iv) Requires immediate resources.

Shareholders Wealth Maximisation	Highest market value of shares.	(i) Emphasizes the long term gains (ii) Recognises risk or uncertainty (iii) Recognises the timing of returns (iv) Considers shareholders' return.	(i) Offers no clear relationship between financial decisions and share price. (ii) Can lead to management anxiety and frustration.
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(b) Various types of Preference shares can be as below:

Sl. No.	Type of Preference Shares	Salient Features
1.	Cumulative	Arrear Dividend will accumulate.
2.	Non-cumulative	No right to arrear dividend.
3.	Redeemable	Redemption should be done.
4.	Participating	Can participate in the surplus which remains after payment to equity shareholders.
5.	Non-Participating	Cannot participate in the surplus after payment of fixed rate of dividend.
6.	Convertible	Option of converting into equity shares.

6B: STRATEGIC MANAGEMENT



QUESTIONS

Multiple Choice Questions

1. Once upon a time in the land of sun, sand, and vibrant cultures, there existed a company named "MuseoGoa" - a company that managed museums in the beautiful state of Goa. MuseoGoa had a vision to celebrate the rich history and culture of Goa, but their journey was not without its fair share of challenges.

MuseoGoa had chosen a picturesque location in a quaint village to build their first museum. However, this initial enthusiasm was met with an uproar from the local communities. The villagers were concerned about the impact on their way of life and traditions. They worried that the influx of tourists might disrupt their peaceful existence.

To address this challenge, MuseoGoa applied Mendelow's matrix, identifying the local communities as key stakeholders. They decided to engage in open dialogues, understanding and respecting the villagers' concerns. MuseoGoa initiated community-building activities, such as involving locals in museum operations, supporting local artisans, and organizing cultural events that showcased the village's heritage. Slowly but steadily, the company transformed from being perceived as a threat to a valued partner within the community.

While MuseoGoa had successfully resolved their initial issues with the local community, they faced another challenge. Their location, although idyllic, was a bit off the beaten path. Tourists typically preferred the bustling beaches closer to the city, and this posed a real challenge. MuseoGoa decided to employ a pricing strategy. They priced their tickets affordably, significantly cheaper than the city's attractions. This strategy attracted budget-conscious tourists who were looking for unique experiences in Goa without burning a hole in their pockets. As

word spread about the cultural gem tucked away in the village, visitors started flocking in, drawn not just by the museum's charm but also the economical ticket prices.

In the age of social media, MuseoGoa knew that word-of-mouth was no longer limited to whispers. They tapped into the power of social media to promote their unique museum experience. MuseoGoa ran interactive campaigns, encouraging visitors to share their experiences on various platforms. One particular Instagram post featuring a vibrant Goan mural in the museum went viral. This was the turning point. The picture-perfect aesthetics of the museum attracted influencers, bloggers, and travel enthusiasts, making MuseoGoa a social media sensation. Visitors came pouring in, not just from India but from across the globe, eager to capture their own moments at the "Instagrammable Museum of Goa."

With success came ambition. MuseoGoa decided to expand its footprint beyond Goa. To guide this expansion, they conducted a strategy audit and trend analysis. They identified emerging cultural and tourism trends and found potential markets in Pune and Trivandrum.

In Pune, MuseoGoa curated a special exhibition that celebrated the fusion of Goan and Maharashtrian cultures. They strategically partnered with local influencers and travel agencies to market the new experience. The expansion into Pune was met with resounding success.

For Trivandrum, MuseoGoa recognized the importance of local traditions and the distinct flavor of Kerala. They tailored their offerings to harmonize with the regional culture. MuseoGoa became the gateway for tourists to explore Kerala's rich heritage, with the museum acting as a bridge between Goa and Kerala's cultural tapestry.

MuseoGoa's journey from initial uproar to expansion was a testament to their commitment to community building, strategic pricing, social media savvy, and a keen eye for trends. The company continued to flourish, celebrating the diverse cultural tapestry of India and making history come alive in every location they touched.

Based on the above Case Scenario, answer the Multiple Choice Questions.

- (i) Which strategic management concept did MuseoGoa use to address the initial concerns of the local community?
 - (a) SWOT analysis
 - (b) Mendelow's matrix
 - (c) Cost leadership strategy
 - (d) Porter's Five Forces model
- (ii) MuseoGoa's idyllic location in a quaint village posed a challenge as tourists preferred beaches closer to the city. To attract visitors, MuseoGoa priced their tickets affordably, cheaper than city attractions, drawing budget-conscious tourists looking for unique experiences. What business strategy did MuseoGoa employ to attract more tourists?
 - (a) Cost leadership strategy
 - (b) Differentiation strategy
 - (c) Focus strategy
 - (d) Diversification strategy
- (iii) How did MuseoGoa approach its expansion into new markets such as Pune and Trivandrum?
 - (a) Outsourcing strategy
 - (b) Franchising strategy
 - (c) Product diversification strategy
 - (d) Market development strategy
- (iv) Which element of the 7S McKinsey model is demonstrated by MuseoGoa's strategic use of social media and pricing strategies to attract visitors?
 - (a) Style
 - (b) Strategy

- (c) Shared Values
- (d) Skills
- (v) What played a crucial role in MuseoGoa's success in Pune and Trivandrum?
 - (a) Strategic partnerships
 - (b) Aggressive advertising
 - (c) Product differentiation
 - (d) Vertical integration
- 2. Jaago Lights, a successful brand from Jalandhar, aimed to enter the Middle East market by teaming up with major industry players. They needed to reorganize internal operations and refine product designs, facing pressure to expand quickly and turbulence in existing operations. What is the primary limitation of strategic management highlighted in the business case?
 - (a) Lack of senior management support
 - (b) Time-consuming and complex nature
 - (c) Inability to adapt to market changes
 - (d) Excessive focus on short-term goals
- 3. A traditional desi ghee company modernized its production and introduced pro-biotic desi ghee, facing initial market doubts. Aggressive marketing campaigns highlighted its benefits, gaining acceptance. During which stage of the product life cycle did the desi ghee company face doubts but gained acceptance through aggressive marketing campaigns?
 - (a) Introduction stage
 - (b) Growth stage
 - (c) Maturity stage
 - (d) Decline stage
- 4. A small tech company focused on enhancing their main product, which became crucial across various industries due to its increased power and

adaptability. Their early partnerships and smart decisions facilitated rapid growth, leading to a \$5 billion valuation in just five years. According to C.K. Prahalad and Gary Hamel, which area represents the tech company's core competency?

- (a) Customer Value
 - (b) Competitor Differentiation
 - (c) Product Differentiation
 - (d) Application to Other Markets
5. A women's clothing brand recognized new opportunities and researched emerging trends and consumer preferences. They introduced a new clothing line, received positive feedback from initial trials, and grew through strategic partnerships and targeted advertising. What strategic choice best describes this approach?
- (a) Product Development
 - (b) Market Development
 - (c) Market Penetration
 - (d) Diversification
6. For over a hundred years, the KDH business has thrived by leveraging strategic control as a cornerstone of its strategic approach. Regular evaluations of goals and performance ensured they stayed responsive to shifting market trends and evolving customer needs. Which type of strategic control is highlighted here?
- (a) Premise Control
 - (b) Special Alert Control
 - (c) Implementation Control - Milestone Reviews
 - (d) Implementation Control - Monitoring Strategic Thrusts

Descriptive Questions**Chapter 1-Introduction to Strategic Management**

7. Tech Innovators Inc., a rapidly expanding technology company, aims to lead in artificial intelligence (AI) and machine learning (ML). With recent growth, the company is evaluating which organizational structure will best support its vision for innovation and leadership in AI technologies. They are considering three options: the Functional and Divisional Relationship for specialization, the Horizontal Relationship for flat, collaborative management, and the Matrix Relationship for cross-functional teams. Which of these relationships—Functional and Divisional, Horizontal, or Matrix—will most effectively achieve Tech Innovators Inc.'s strategic goals, and why?
8. Strategic management helps an organization to work through changes in the environment to gain competitive advantage. In light of statement discuss its benefits.

Chapter 2-Strategic Analysis: External Environment

9. A company has recently launched a new product in the market. Initially, it faced slow sales growth, limited markets, and high prices. However, over time, the demand for the product expanded rapidly, prices fell, and competition increased. Identify the stages of the product life cycle (PLC) that the company went through.
10. Rajiv Arya owns an electrical appliance company specializing in the manufacture of domestic vacuum cleaners. The market is competitive, with four other manufacturers offering similar products and achieving comparable sales volumes. Additionally, these rival firms hold several patents related to the vacuum cleaner technology. The supplier base for raw materials is extensive, with multiple suppliers available. Identify and explain the significant forces from Porter's Five Forces framework that are relevant to Rajiv Arya's company.

Chapter 3-Strategic Analysis: Internal Environment

11. In spite of high commodity inflation, shortage of components and the threat of third wave of COVID-19 pandemic in India, manufacturers of packaged goods, home appliances and consumer electronics are

expecting the business to grow by 12 to 25 percent in the coming months. After one-and-a-half years of disruption, manufacturers are now confident about managing their inventories better, keeping their supply channels well-stocked and preparing themselves to minimize the impact of any COVID related restrictions even as they gear up for the festive season, which usually accounts for 25 to 35 percent of their yearly sales.

The home appliances sector could be an example. After a dismal April-June quarter in the year 2021, producers of air conditioners, refrigerators and washing machines are expecting their business to grow by 15-20 percent in the months to come. All the companies operating in the sector have geared up to grab the opportunities available in the market.

A leading company in the home appliances domain, XXP India, is planning to launch various innovative product designs and offer loyalty programmes to lure consumers.

With reference to Michael Porter's generic strategies, identify which strategy XXP India has planned for? Explain how this strategy will be advantageous to the company to remain profitable?

12. How can Mendelow's Matrix be used to analyze and manage the stakeholders effectively?

Chapter 4-Strategic Choices

13. Pizza Galleria was India's first pizza delivery chain enjoying monopoly for several years. However, after the entry of Modino and Uncle Jack it is struggling to compete. Both Modino and Uncle Jack have opened several eateries and priced the product aggressively. In the last four years the chain has suffered significant losses. The chain wishes to know whether they should go for a turnaround strategy. List out components of action plan for turnaround strategy.
14. Distinguish between Concentric Diversification and Conglomerate Diversification. "

Chapter 5-Strategy Implementation and Evaluation

15. A Mumbai-based conglomerate, PQR Ltd., has announced a major restructuring of its business operations. The company has decided to split its business into four separate units: Manufacturing, Retail, Services, and Technology. Each unit will operate as a separate business, with delegated responsibility for day-to-day operations and strategy to the respective unit managers. Identify the organization structure that PQR Ltd. has planned to implement. Discuss any four attributes and the benefits the firm may derive by using this organization structure.
16. Why Strategic Performance Measures are essential for organizations?


SUGGESTED ANSWERS/HINTS

MCQ No.	Answer	
1.	(i)	(b)
	(ii)	(a)
	(iii)	(d)
	(iv)	(b)
	(v)	(a)
2.		(b)
3.		(a)
4.		(d)
5.		(a)
6.		(c)

7. The Matrix Relationship is the most effective structure for Tech Innovators Inc. to achieve its vision of leadership in AI technologies. This structure promotes cross-functional collaboration, essential for managing complex AI projects and fostering innovation. By integrating expertise from various departments into temporary, task-based teams,

the Matrix Relationship supports dynamic project management and aligns well with the company's strategic goals for advancing AI technologies. Despite its complexity, this approach provides the flexibility and collaboration necessary for a leading-edge AI and ML focus.

Relationship	Benefits	Drawbacks	Suitability for AI Leadership
Functional and Divisional	Specialization, clear management of functions and products.	Potential for departmental isolation, limited collaboration.	Less effective for cross-functional AI projects.
Horizontal	Open communication, encourages innovation and fast idea sharing.	Hard to scale, unclear roles and responsibilities.	Suitable for startups, less for large AI initiatives.
Matrix	Facilitates cross-functional collaboration, flexible resource management for complex projects.	Complex reporting structures, potential conflicts.	Ideal for managing diverse, innovative AI projects.

8. Strategic management involves developing the company's vision, environmental scanning, strategy formulation, implementation, evaluation and control. It emphasizes the monitoring and evaluation of external opportunities and threats in the light of a company's strengths and weaknesses and designing strategies for survival and growth. It helps in the creation of a competitive advantage to outperform the competitors and also guides the company successfully through all changes in the environment.

The major benefits of strategic management are:

- ◆ Strategic management gives directions to the company to move ahead. It defines the goals and mission.
- ◆ It helps organisations to be proactive instead of reactive in shaping their future.
- ◆ It provides frameworks for all major decisions of an enterprise such as decisions on businesses, products, markets, manufacturing facilities, investments and organisational structure. It provides better guidance to the entire organisation on the crucial point - what it is trying to do.
- ◆ It helps organisations to identify the available opportunities and identify ways and means to achieve them.
- ◆ It serves as a corporate defence mechanism against mistakes and pitfalls.
- ◆ It helps to enhance the longevity of the business.
- ◆ It helps the organisation to develop certain core competencies and competitive advantages that would facilitate survival and growth.

9. The company went through the following stages of the product life cycle (PLC):

Introduction stage: Initially, the company faced slow sales growth, limited markets, and high prices, which are characteristic of the introduction stage. During this stage, competition is almost negligible, and customers have limited knowledge about the product.

Growth stage: Over time, the demand for the product expanded rapidly, prices fell, and competition increased. These are typical features of the growth stage in the PLC. In this stage, the product gains market acceptance, and customers become more aware of the product's benefits and show interest in purchasing it.

10. The competitive rivalry will be a significant force in case of company of Rajiv Arya as all the rivals are similar in sizes and are manufacturing similar products. It is difficult for any single manufacturer to dominate

the market. Large number of patents will make it difficult for new entrants to break into the market. Further, as there are a large number of small suppliers the power that suppliers can exert will also be low.

There is no information relating to substitutes and bargaining power of customers in the information given in scenario. However, a domestic vacuum cleaner will directly compete with other options such as house maids. Availability of house maids at low cost can significantly disturb the sales of products.

Further, as the products are similar customers can easily shift from one company to another. This will only enhance competitive rivalry.

11. According to Michael Porter, strategies allow organizations to gain competitive advantage from three different bases: cost leadership, differentiation, and focus. Porter called these base generic strategies.

XXP India Ltd. has planned for Differentiation Strategy. The company is planning to launch various innovative product designs and offer loyalty programmes to lure customers.

Differentiation strategy should be pursued only after a careful study of buyers' needs and preferences to determine the feasibility of incorporating one or more differentiating features into a unique product that features the desired attributes. A successful differentiation strategy allows a firm to charge a higher price for its product and to gain customer loyalty, because consumers may become strongly attached to the differentiated features.

Advantages of Differentiation Strategy

A differentiation strategy may help an organisation to remain profitable even with rivalry, new entrants, suppliers' power, substitute products, and buyers' power.

1. **Rivalry** - Brand loyalty acts as a safeguard against competitors. It means that customers will be less sensitive to price increases, as long as the firm can satisfy the needs of its customers.
2. **Buyers** - They do not negotiate for price as they get special features, and they have fewer options in the market.

3. **Suppliers** – Because differentiators charge a premium price, they can afford to absorb higher costs of supplies as the customers are willing to pay extra too.
 4. **Entrants** – Innovative features are an expensive offer. So, new entrants generally avoid these features because it is tough for them to provide the same product with special features at a comparable price.
 5. **Substitutes** – Substitute products can't replace differentiated products which have high brand value and enjoy customer loyalty.
12. Mendelow's Matrix can be used effectively to analyze and manage stakeholders through a grid-based approach by the following steps:
1. **Identify Stakeholders:** Begin by identifying all relevant stakeholders for your project or organization. This includes individuals, groups, or organizations that may be impacted by or have an impact on your activities.
 2. **Assess Power and Interest:** For each stakeholder, assess their power to influence your project or organization and their level of interest in its success. Power can be assessed based on factors such as authority, resources, and expertise, while interest can be gauged by their level of involvement, expectations, and potential benefits or risks.
 3. **Plot Stakeholders on the Grid:** Create a grid with Power on one axis and Interest on the other. Plot each stakeholder on the grid based on your assessment. Stakeholders with high power and high interest are placed in the "Key Players" quadrant, those with high power but low interest are in the "Keep Satisfied" quadrant, those with low power but high interest are in the "Keep Informed" quadrant, and those with low power and low interest are in the "Low Priority" quadrant.



4. **Develop Strategies for each Quadrant:** Based on the placement of stakeholders in the grid, develop specific strategies for managing each quadrant:

- **Key Players:** Fully engage with these stakeholders, seek their input, and keep them informed. They are crucial for the success of your project, so their needs and expectations should be a top priority.
- **Keep Satisfied:** These stakeholders have significant power but may not be as interested in your project. Keep them satisfied by providing regular updates and addressing any concerns they may have to prevent them from becoming detractors.

- **Keep Informed:** While these stakeholders may not have much power, they are highly interested in your project. Keep them informed to ensure they remain supportive and to leverage their insights and feedback.
- **Low Priority:** These stakeholders have low power and interest. Monitor them for any changes but allocate minimal resources to managing their expectations.

5. **Monitor and Adapt:** Continuously monitor the power and interest of stakeholders and adjust your strategies accordingly. Stakeholders may move between quadrants based on changing circumstances, so it's important to remain flexible and responsive.

By using Mendelow's Matrix as a grid-based tool, you can effectively analyze and manage stakeholders by tailoring your engagement strategies to their specific needs and expectations, ultimately increasing the likelihood of project success.

13. Pizza Chain may choose to have turnaround strategy if there are:

- ◆ Persistent negative cash flow from business.
- ◆ Uncompetitive products or services.
- ◆ Declining market share.
- ◆ Deterioration in physical facilities.
- ◆ Over-staffing, high turnover of employees, and low morale.
- ◆ Mismanagement.

For turnaround strategies to be successful, it is imperative to focus on the short and long-term financing needs as well as on strategic issues. The chain may attempt to leverage the potential Indian market by engaging a new logistics partner. It may bring innovation in food items, as well as quality and improvements in the overall dine-in and delivery experience. During the turnaround, the "product mix" may be changed, requiring the organization to do some repositioning.

A workable action plan for turnaround would involve:

Stage One – Assessment of current problems: The first step is to assess the current problems and get to the root causes and the extent of damage the problem has caused.

Stage Two – Analyze the situation and develop a strategic plan: Before making any major changes; determine the chances of the business's survival. Identify appropriate strategies and develop a preliminary action plan.

Stage Three – Implementing an emergency action plan: If the organization is in a critical stage, an appropriate action plan must be developed to stop the bleeding and enable the organization to survive. A positive operating cash flow must be established as quickly as possible and enough funds to implement the turnaround strategies must be raised.

Stage Four – Restructuring the business: The financial state of the organization's core business is particularly important. If the core business is irreparably damaged, then the outlook for the entire organization may be bleak. Efforts to be made to position the organization for rapid improvement.

Stage Five – Returning to normal: In the final stage of turnaround strategy process, the organization should begin to show signs of profitability, return on investments and enhancing economic value-added. Emphasis is placed on a number of strategic efforts such as carefully adding new products and improving customer service, creating alliances with other organizations, increasing the market share, etc.

14. The following are the principal points of distinction between concentric diversification and conglomerate diversification:
- (i) Concentric diversification occurs when a firm adds related products or markets. On the other hand, conglomerate diversification occurs when a firm diversifies into areas that are unrelated to its current line of business.
 - (ii) In concentric diversification, the new business is linked to the existing businesses through process, technology or marketing. In

conglomerate diversification, no such linkages exist; the new business/product is disjointed from the existing businesses/products.

- (iii) The most common reasons for pursuing concentric diversification are that opportunities in a firm's existing line of business are available. However, common reasons for pursuing a conglomerate growth strategy are that opportunities in a firm's current line of business are limited or opportunities outside are highly lucrative.

- 15.** PQR Ltd. has planned to implement the Strategic Business Unit (SBU) structure. Very large organisations, particularly those running into several products, or operating at distant geographical locations that are extremely diverse in terms of environmental factors, can be better managed by creating strategic business units. SBU structure becomes imperative in an organisation with increase in number, size and diversity.

The attributes of an SBU and the benefits a firm may derive by using the SBU Structure are as follows:

- ◆ A scientific method of grouping the businesses of a multi – business corporation which helps the firm in strategic planning.
- ◆ An improvement over the territorial grouping of businesses and strategic planning based on territorial units.
- ◆ Strategic planning for SBU is distinct from rest of businesses. Products/ businesses within an SBU receive same strategic planning treatment and priorities.
- ◆ Each SBU will have its own distinct set of competitors and its own distinct strategy.
- ◆ The CEO of SBU will be responsible for strategic planning for SBU and its profit performance.
- ◆ Products/businesses that are related from the standpoint of function are assembled together as a distinct SBU.
- ◆ Unrelated products/ businesses in any group are separated into separate SBUs.

- ◆ Grouping the businesses on SBU lines helps in strategic planning by removing the vagueness and confusion.
 - ◆ Each SBU is a separate business and will be distinct from one another on the basis of mission, objectives etc.
16. Strategic performance measures are essential for organizations for several reasons:
- ◆ **Goal Alignment:** Strategic performance measures help organizations align their strategies with their goals and objectives, ensuring that they are on track to achieve their desired outcomes.
 - ◆ **Resource Allocation:** Strategic performance measures provide organizations with the information they need to make informed decisions about resource allocation, enabling them to prioritize their efforts and allocate resources to the areas that will have the greatest impact on their performance.
 - ◆ **Continuous Improvement:** Strategic performance measures provide organizations with a framework for continuous improvement, enabling them to track their progress and make adjustments to improve their performance over time.
 - ◆ **External Accountability:** Strategic performance measures help organizations demonstrate accountability to stakeholders, including shareholders, customers, and regulatory bodies, by providing a clear and transparent picture of their performance.